ONLINE SUPPLEMENTARY MATERIALS

Targeted systemic therapies for psoriatic arthritis: a systematic review and comparative synthesis of short-term articular, dermatological, enthesitis and dactylitis outcomes

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Supplementary Methods

Table S 1 Search terms – Embase

#	Searches	Results
1	psoriatic arthritis/	22489
2	(psoria* adj5 (arthr* or polyarthr* or poly-arthr* or oligoarthr* or oligo-arthr* or rheumat*)).tw,kw.	21763
3	1 or 2	27248
4	brodalumab/ or (brodalumab or kyntheum).mp.	1081
5	(Risankizumab or BI-655066 or ABBV-066 or BI655066 or ABBV066).mp.	405
6	(Netakimab or BCD-085).mp.	24
7	Tildrakizumab.mp.	441
8	Filgotinib.mp.	391
9	(Upadacitinib or ABT-494 or ABT494).mp.	432
10	(Bimekizumab or UCB4940).mp.	102
11	secukinumab/ or (secukinumab or cosentyx).mp.	3669
12	certolizumab pegol/ or (Certolizumab or Cimzia).mp.	7227
13	golimumab/ or (golimumab or simponi).mp.	7064
14	apremilast/ or (apremilast or otezla or otezia).mp.	2099
15	ustekinumab/ or (ustekinumab or stelara).mp.	7270
16	etanercept/ or (etanercept or enbrel).mp.	32244
17	infliximab/ or (infliximab or inflectra or remsima or remicade).mp.	51870
18	adalimumab/ or (adalimumab or humira).mp.	34305
19	tofacitinib/ or (tofacitanib or Xeljanz).mp.	4375
20	ixekizumab/ or (ixekizumab or talz).mp.	1765
21	(guselkumab or tremfya).mp. or guselkumab/	726
22	(abatacept or orencia).mp. or abatacept/	9443
23	(janus kinase inhibitor* or JAK1 or JAK inhibitor*).tw,kw.	8488
24	(phosphodiesterase iv inhibitor* or PDE4 inhibitor*).tw,kw.	2032
25	(interleukin inhibitor or IL-17* or IL17 or interleukin-17* or IL-23* or IL23 or interleukin-23*).tw,kw.	47391
26	(tumo?r necrosis factor inhibitor* or TNF inhibitor* or tumo?r necrosis factor antagonist* or TNF antagonist*).tw,kw.	8585
27	or/4-26	143959
28	randomized controlled trial/ or randomized controlled trials as topic/	733085
29	(randomi?ed control* or rct).ti.	111779
30	randomization/	87617
31	Single-Blind Method/ or single blind.mp.	48164
32	double blind method/ or double blind.mp.	252824
33	(phase 2 or phase II or phase III or phase 3 or phase IV or phase 4).mp.	278460
34	((singl$ or doubl$ or treb$ or tripl$) adj (blind$3 or mask$3)).mp.	309493
35	Random Allocation/	83784
36	(allocat* adj3 random*).mp.	45459
37	placebo$.mp.	457598
38	Prospective Studies/	513595
39	(prospective adj (trial or study)).mp.	688656

40	Controlled Clinical Trial/	464393
41	or/28-40	2059304
42	Case Report/	2508958
43	case stud*.hw.	71425
44	case report?.hw.	2508982
45	Letter/	1069988
46	Editorial/	646930
47	or/42-46	4070371
48	41 not 47	1978997
49	(202008* or 202007* or 202006* or 202005* or 202004* or 202003*).dd.	425295
50	3 and 27 and 48	3385
51	50 and 49	29

Table S 2 Search strategy

Search type	Resources accessed	Years searched	Date accessed
Electronic databases	Accessed via Ovid: • Embase • Ovid MEDLINE In-Process Accessed via the Cochrane Library Website: • The Cochrane Library	No limits applied	11 March 2020; Updated 18 August 2020
Congress abstract search	• American College of Rheumatology (ACR) • British Society for Rheumatology (BSR) • European League Against Rheumatism (EULAR)	2017-2019 2020 2020	1 May 2020 1 May 2020 25 August 2020
Clinical trial databases	• ClinicalTrials.gov • The WHO International Clinical Trials Registry Platform (ICTRP) • European Union Drug Regulating Authorities Clinical Trials Database (EudraCT)	No limits applied	1 May 2020; Updated 25 August 2020
Bibliographic search	All bibliographies of SLR/NMAs identified for inclusion at full-text review stage were reviewed for additional publications not identified in electronic or congress searches	Not applicable	13 May 2020; Updated 24 August 2020

Table S 3 Eligibility criteria - PICOS

Patient population	**Include** • Patients ≥ 16 years-old with active psoriatic arthritis **Exclude** • Paediatric patients (< 16 years-old) • Trials consisting exclusively of DMARD-naïve patients • Patients with psoriatic arthritis included in randomised studies intended to evaluate therapies primarily for the treatment of plaque psoriasis • Animal/in vitro studies
Interventions & Comparators	**Intervention** • Abatacept • Apremilast • Adalimumab • Bimekizumab • Brodalumab • Certolizumab pegol • Etanercept • Filgotinib • Golimumab • Guselkumab • Infliximab • Ixekizumab • Netakimab • Risankizumab • Secukinumab • Tildrakizumab • Tofacitinib • Upadacitinib • Ustekinumab **Comparator** • Any comparator Interventions and comparators can be reported either as a monotherapy or in combination with another systemic treatment.
Outcomes	• Skin: o Psoriasis Area and Severity Index (PASI), either as a continuous measure (e.g. change from baseline) or as a dichotomous outcome (e.g. PASI75, PASI90 and PASI100) • Disability and physical function: o Disease Activity Score for 28-joint count (DAS28) o Health Assessment Questionnaire–Disability Index (HAQ-DI) • Disease activity:

	○ Bath Ankylosing Spondylitis Disease Activity Index (BASDAI) ○ ACR score ○ Psoriatic Arthritis Response Criteria (PsARC) ○ Clinical Disease Activity Index (CDAI) ○ Psoriatic Arthritis Disease Activity Score (PASDAS) ○ Minimal Disease Activity (MDA) ○ Group for Research and Assessment of Psoriasis and Psoriatic Arthritis (GRAPPA) Composite Score (GRACE) ○ Composite Psoriatic Disease Activity Index (CPDAI) ○ Disease Activity in Psoriatic Arthritis (DAPSA) ○ Enthesitis resolution ○ Dactylitis resolution • Health related quality of life: ○ Psoriasis Symptom Inventory (PSI) ○ EQ-5D ○ 36-Item Short Form Health Survey (SF-36) ○ Psoriatic Arthritis Quality of Life (PsAQoL) questionnaire • Adverse events
Study type	Include • RCTs (if ≥50 patients were randomised to at least one trial arm), regardless of phase • SLRs (and NMAs/ITCs) of RCTs[1] • Pooled analyses of RCTs[2] Exclude • RCTs (if all trial arms contain fewer than 50 patients), observational studies, non-randomised studies, single-arm studies, extensions of RCTs that do not preserve randomisation, registry studies, case reports, case studies, editorials, commentaries
Language	English

Abbreviations: ACR, American College of Rheumatology; DMARD, disease-modifying antirheumatic drug; EQ-5D, EuroQol- 5 Dimension; ITC, indirect treatment comparison; NMA, network meta-analysis; RCT, randomised controlled trial; SLR, systematic literature review.

[1] Bibliographies of included SLRs and indirect comparison publications are reviewed to determine whether they contain any relevant RCTs that meet the inclusion criteria. The results and data used in the SLRs and indirect comparisons themselves are not of interest.

[2] Bibliographies of included pooled analyses are reviewed to determine whether they contain relevant RCTs that meet the inclusion criteria. The results and data used in the pooled analyses are only of interest if data are not available for the individual RCTs reported in the pooled analysis publications.

Supplementary Results

Table S 4 Study design and baseline characteristics of trials included in the NMA

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naive	bDMARD-exposed											
ASTRAEA[1]	✓	✓	✓	✓	✓	✓	✓	CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; Active plaque PsO with ≥1 qualifying target lesion ≥2 cm in diameter	Placebo (211)	8.8 ± 8.3	49.8 ± 11.3	53.1	61.6[2d]	100
									Abatacept 125 mg (213)	8.3 ± 8.1	51 ± 10.7	56.8	60.6[2d]	100
ADEPT[2]	✓	✓		✓			✓	≥3 swollen joints and ≥3 tender joints or painful joints; either active psoriatic skin lesions or a documented history of PsO	Placebo (162)	9.2 ± 8.7	49.2 ± 11.1	45.1	0[2d]	NR
									Adalimumab 40 mg Q2W (153)	9.8 ± 8.3	48.6 ± 12.5	43.7	0[2d]	NR
Genovese 2007[3]	✓	✓					✓	≥3 swollen joints and ≥3 tender joints; either an active psoriatic skin lesion of chronic plaque PsO or a documented history of chronic plaque PsO diagnosed by the investigator or a dermatologist	Placebo (51)	7.2 ± 7	47.7 ± 11.3	49	0[2d]	100
									Adalimumab 40 mg Q2W (51)	7.5 ± 7	50.4 ± 11	43.1	0[2d]	100
Mease 2018[4]	✓			✓			✓	≥3 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; ≥1 psoriatic skin lesion ≥2 cm in diameter	Placebo (24)	7.6 ± 7.2	47.7 ± 13.7	50	NR	NR
									Adalimumab 40 mg Q2W (72)	8.4 ± 9.2	50.5 ± 12	45.8	NR	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
									ABT-122 120 mg (71)	5.9 ± 7.1	51 ± 12.4	52.1	NR	NR
									ABT-122 240 mg (73)	7.5 ± 8.2	47.4 ± 13.8	50.7	NR	NR
ACTIVE[5]	✓	✓			✓		✓	≥3 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; CRP ≥0.2 mg/dL	Placebo (109)	3.6 ± 5.5	48 ± 13.8	59.6	0[2c]	71.6
									Apremilast 30 mg BID (110)	4 ± 4.5	50.7 ± 12.2	52.7	0[2c]	67.3
Schett 2012[6]	✓						✓	≥6 months diagnosis; Moll & Wright criteria; ≥3 swollen joints and ≥3 tender joints; RF-	Placebo (68)	7.3 ± NR	51.1 ± NR	52.9	NR	NR
									Apremilast 20 mg BID (69)	8.4 ± NR	50.9 ± NR	37.7	NR	NR
									Apremilast 40 mg QD (67)	7.6 ± NR	49.9 ± NR	52.2	NR	NR
PALACE 1[7]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (168)	7.3 ± 7.1	51.1 ± 12.13	47.6	24.4[2d]	71.4
									Apremilast 20 mg BID (168)	7.2 ± 6.8	48.7 ± 10.99	49.4	22[2d]	76.8
									Apremilast 30 mg BID (168)	8.1 ± 8.1	51.4 ± 11.72	54.8	24.4[2d]	73.8
PALACE 2[8]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (159)	7.8 ± 8.3	51.2 ± 11	53.5	14.5	84.9
									Apremilast 20 mg BID (163)	7.8 ± 8.6	50.9 ± 11.8	58.3	17.2	82.8
									Apremilast 30 mg BID (162)	6.8 ± 7.6	50.5 ± 11.2	58.6	14.2	82.7

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
PALACE 3[9]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; active PsO with ≥1 plaque PsO skin lesion ≥2 cm in size	Placebo (169)	6.8 ± 6.5	49.5 ± 11.6	54	28	72
									Apremilast 20 mg BID (169)	7.7 ± 7.7	49.6 ± 12.1	53	30	70
									Apremilast 30 mg BID (167)	7.5 ± 7.6	49.9 ± 11.4	53	26	74
Mease 2014[10]	✓						✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (55)	8.4 ± 7.5	53 ± 13	55	45[2b]	42[2a]
									Brodalumab 140 mg (57)	9.4 ± 7.5	53 ± 10	65	53[2b]	54[2a]
									Brodalumab 280 mg (56)	8.1 ± 7.9	51 ± 12	71	55[2b]	54[2a]
AMVISION 1[11]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints (excluding the distal interphalangeal joints); ≥1 psoriatic skin lesion	Placebo (161)	9.4 ± 9.3	48.1 ± 11.79	49.7	27.3[2c]	NR
									Brodalumab 140 mg (158)	8.1 ± 8.1	49.9 ± 12.8	50.6	26.6[2c]	NR
									Brodalumab 210 mg (159)	8.2 ± 8.2	49.1 ± 12.24	44	28.9[2c]	NR
AMVISION 2[11]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; ≥1 psoriatic skin lesion	Placebo (161)	6.4 ± 7.7	48.3 ± 13	52.8	36[2c]	NR
									Brodalumab 140 mg (160)	6.5 ± 7.4	47.4 ± 12.8	50	33.8[2c]	NR
									Brodalumab 210 mg (163)	7.1 ± 7.5	47 ± 12.6	51.5	34.4[2c]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
RAPID-PsA[12]	✓	✓	✓	✓			✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; either ESR ≥28 mm/h (Westergren) or CRP >upper limit of normal (7.9 mg/L), and have active psoriatic skin lesions or a documented history of PsO	Placebo (136)	7.9 ± 7.7	47.3 ± 11.1	58.1	19.1[2d]	NR
									Certolizumab pegol 200 mg Q2W (138)	9.6 ± 8.5	48.2 ± 12.3	53.6	22.5[2d]	NR
									Certolizumab pegol 400 mg Q4W (135)	8.1 ± 8.3	47.1 ± 10.8	54.1	17[2d]	NR
Mease 2004[13]	✓	✓		✓			✓	Moll & Wright criteria; ≥3 swollen joints and ≥3 tender joints; ≥1 clinical subtype of PsA (DIP joint involvement, polyarticular arthritis, arthritis mutilans, asymmetric peripheral arthritis, or ankylosing spondylitis-like arthritis); stable plaque PsO with skin lesion ≥2 cm in diameter	Placebo (104)	9.2 ± NR	47.3 ± NR	55	0[2d]	NR
									Etanercept 25 mg BIW (101)	9 ± NR	47.6 ± NR	43	0[2d]	NR
PRESTA[14]	✓	✓		✓			✓	≥3 months diagnosis; ≥2 swollen joints and ≥2 tender/painful joints;	Etanercept 50 mg QW (373)	7 ± 7	47 ± 11	38.3	0[2d]	40[2a]

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
								moderate-severe PsA according to PGA; active but clinically stable PsO involving ≥10% of the total BSA and a PGA of PsO of moderate-to-severe;RF- within for ≥6 months before screening	Etanercept 50 mg BIW (379)	7 ± 7	46 ± 11	35.9	0[2d]	32[2a]
SEAM-PsA[15]	✓	✓			✓	✓	✓	CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; skin lesion ≥2 cm in diameter	Placebo + methotrexate 20 mg QW (284)	3.6 ± 6.8	48.7 ± 13.1	56.3	0[2c]	13.4
									Placebo + etanercept 50 mg QW (284)	3.1 ± 6	48.5 ± 13.5	46.8	0[2c]	9.2
									Etanercept 50 mg + methotrexate 20 mg QW (283)	3 ± 6	48.1 ± 12.7	49.1	0[2c]	15.2
EQUATOR[16]	✓	✓	✓	✓	✓	✓	✓	≥12 weeks diagnosis; CASPAR criteria; ≥5 swollen joints and ≥5 tender joints; history of/or active plaque PsO	Placebo (66)	7 ± 6.2	50 ± 10.9	45	14[2d]	100[2e]
									Filgotinib 200 mg (65)	7 ± 6.7	49 ± 12.2	55	17[2d]	100[2e]
GO-REVEAL[17]	✓	✓		✓			✓	≥6 months diagnosis; ≥3 swollen and ≥3 tender joints;	Placebo (113)	7.6 ± 7.9	47 ± 10.6	38.9	0[2d]	67[2a]

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
								≥1 subset of PsA and the presence of plaque PsO with a qualifying lesion ≥2 cm in diameter; RF-	Golimumab 50 mg (146)	7.2 ± 6.8	45.7 ± 10.7	39	0[2d]	66[2a]
									Golimumab 100 mg (146)	7.7 ± 7.8	48.2 ± 10.9	41.1	0[2d]	73[2a]
GO-VIBRANT[18]	✓	✓		✓			✓	≥6 months diagnosis; CASPAR criteria; ≥5 swollen and ≥5 tender; high-sensitivity CRP level of ≥0.6 mg/dl at screening despite current or previous DMARD therapy and/or NSAID therapy, or demonstrate intolerance to these agents	Placebo (239)	5.3 ± 5.9	46.7 ± 12.5	49.4	0[2c]	NR
									Golimumab 2 mg/kg (241)	6.2 ± 6	45.7 ± 11.3	46.9	0[2c]	NR
GO-DACT[19][5]						✓		≥3 months diagnosis; CASPAR criteria; ≥1 tender dactylitis, refractory to ≥2 systemic NSAIDs, at optimal dosage, for 3 months and ≥1 other site of active inflammation (peripheral joints, enthesis, spine, skin or nails)	Placebo + methotrexate 15-25 mg (23)	25.9 ± 5.4	44.1 ± 24.6	NR	0	NR
									Golimumab 50 mg + methotrexate 15-25 mg (21)	29 ± 4.5	46.2 ± 15.5	NR	0	NR
DISCOVER 1[20]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen and ≥3	Placebo (126)	7.2 ± 7.6	49 ± 11.1	52	31[2d]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
								tender joints; CRP concentration of ≥0.3 mg/dL; ≥1 PsA subset: DIP joint involvement, polyarticular arthritis with absence of rheumatoid nodules, arthritis mutilans, asymmetric peripheral arthritis, or spondylitis with peripheral arthritis	Guselkumab 100 mg Q8W (128)	6.4 ± 5.9	48.9 ± 11.5	46	32[2d]	NR
									Guselkumab 100 mg Q4W (128)	6.6 ± 6.3	47.4 ± 11.6	48	30[2d]	NR
DISCOVER 2[21]	✓	✓		✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen and ≥3 tender joints; ≥5 swollen and ≥5 tender joints; CRP concentration of ≥0.6 mg/dL; current or documented history of PsO; ≥1 PsA subset: DIP joint involvement, polyarticular arthritis with absence of rheumatoid nodules, arthritis mutilans, asymmetric peripheral arthritis, or spondylitis with peripheral arthritis	Placebo (247)	5.8 ± 5.6	46.3 ± 11.7	52	0[2c]	NR
									Guselkumab 100 mg Q8W (248)	5.1 ± 5.5	44.9 ± 11.9	48	0[2c]	NR
									Guselkumab 100 mg Q4W (246)	5.5 ± 5.9	45.9 ± 11.5	42	0[2c]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
Deodhar 2018[22]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen and ≥3 tender or painful joints; CRP concentration of ≥0.3 mg/dL; ≥3% of BSA affected by plaque PsO	Placebo (49)	6.9 ± 7.2	44.2 ± 12.4	51	8[2d]	84
									Guselkumab 100 mg Q8W (100)	7 ± 7.2	47.4 ± 12.8	48	9[2d]	90
IMPACT[23]	✓	✓		✓			✓	≥6 months diagnosis; ≥5 swollen and ≥5 tender joints; active peripheral polyarticular arthritis in conjunction with ≥1 of the following criteria: ESR ≥28 mm/h, CRP level ≥15 mg/L, and/or morning stiffness lasting ≥45 minutes	Placebo (52)	11 ± 6.6	45.2 ± 9.7	42.3	0	100
									Infliximab 5mg/kg (52)	11.7 ± 9.8	45.7 ± 11.1	42.3	0	100
IMPACT 2[24]	✓	✓		✓			✓	≥6 months diagnosis; ≥5 swollen and ≥5 tender joints; either CRP levels of ≥15 mg/L and/or morning stiffness lasting ≥45 minutes; active plaque PsO with ≥1 qualifying target lesion ≥2 cm in diameter; RF-	Placebo (100)	7.5 ± 7.8	46.5 ± 11.3	49	0[2d]	NR
									Infliximab 5 mg/kg (100)	8.4 ± 7.2	47.1 ± 12.8	29	0[2d]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
RESPOND[25]	✓	✓		✓			✓	≥3 months diagnosis; ≥5 swollen and ≥5 tender joints; ≥1 of the following: ESR ≥28 mm/h; CRP ≥15 mg/L; or morning stiffness lasting ≥45 minutes; PsO in combination with peripheral articular disease with ≥1 of the following for ≥3 months before screening: DIP joint involvement; polyarticular arthritis in the absence of rheumatoid nodules; arthritis mutilans; or asymmetric peripheral arthritis; RF-	Infliximab 5mg/kg + methotrexate 15 mg (57)	2.8 ± 2.6	40.1 ± 12.3	51.8	0[2c]	22.8
									Methotrexate 15mg (58)	3.7 ± 2.7	42.3 ± 10.5	38.9	0[2c]	15.5
SPIRIT H2H[26]	✓	✓		✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen and ≥3 tender joints; Active psoriatic skin lesions of plaque PsO with a BSA ≥3%; RF-	Adalimumab 40 mg Q2W (283)	5.9 ± 6.4	48.3 ± 12.3	47	0[2c]	100[2e]
									Ixekizumab 80 mg (283)	6.6 ± 7.4	47.5 ± 12	43	0[2c]	100[2e]
SPIRIT P1[27]	✓	✓		✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; either ≥1	Placebo (106)	6.3 ± 6.9	50.6 ± 12.3	54.7	0	22.6[2e]

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
								PsA-related hand or foot joint erosion on centrally read X-rays or CRP >6 mg/L; RF-	Adalimumab 40 mg Q2W (101)	6.9 ± 7.5	48.6 ± 12.4	49.5	0	19.8[2e]
									Ixekizumab 80 mg Q4W (107)	6.2 ± 6.4	49.1 ± 10.1	57.9	0	20.6[2e]
									Ixekizumab 80 mg Q2W (103)	7.2 ± 8	49.8 ± 12.6	53.4	0	22.3[2e]
SPIRIT P2[28]	✓		✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; RF- Have active psoriatic skin lesions or a documented history of plaque PsO	Placebo (118)	9.2 ± 7.3	51.5 ± 10.4	53	100[2d]	100[2e]
									Ixekizumab 80 mg Q4W (122)	11 ± 9.6	52.6 ± 13.6	48	100[2d]	100[2e]
									Ixekizumab 80 mg Q2W (123)	9.9 ± 7.4	51.7 ± 11.9	59	100[2d]	100[2e]
PATERA[29]	✓			✓				6 months diagnosis; CASPAR criteria; 3 tender and 3 swollen joints; ≥1 psoriatic plaque ≥2 cm and/or psoriatic nails and/or history of confirmed plaque PsO	Placebo (97)	5.7 ± 6.5[6]	43.1 ± 11.88	48.45	17.53[2d]	SSZ: 11.34; LEF: 8.25
									Netakimab 120 mg (97)	5.3 ± 6.09[6]	44 ± 11.66	46.39	22.68[2d]	SSZ: 9.28; LEF: 4.12
CHOICE[30]	✓	✓		✓	✓	✓	✓		Placebo 52)	NR ± NR	NR ± NR	NR	0	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
								≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; Target skin psoriatic lesion and a PASI score of ≥1	Secukinumab 150 mg (SC LD) (103)	NR ± NR	NR ± NR	NR	0	NR
									Secukinumab 300 mg (SC LD) (103)	NR ± NR	NR ± NR	NR	0	NR
FUTURE 1[31]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis;CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; Active plaque PsO, with ≥1 psoriatic plaque of ≥2 cm diameter (but not in intertriginous areas such as armpits, or chest between breasts, or groin) or nail changes consistent with PsO or a documented history of plaque PsO	Placebo (202)	NR ± NR	48.5 ± 11.2	52.5	29.2[2d]	NR
									Secukinumab 75 mg (IV LD) (202)	7.66 ± 8.31	48.8 ± 12.2	58.4	29.7[2d]	NR
									Secukinumab 150 mg (IV LD) (202)	8.19 ± 8.33	49.6 ± 11.8	52.5	29.2[2d]	NR
FUTURE 2[32]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; active plaque PsO, nail changes consistent with PsO, or a documented history of plaque PsO	Placebo (98)	7.3 ± 7.8	49.9 ± 12.5	60	35[2d]	NR
									Secukinumab 75 mg (SC LD) (99)	NR ± NR	48.6 ± 11.4	53	34[2d]	NR
									Secukinumab 150 mg (SC LD) (100)	6.5 ± 8.2	46.5 ± 11.7	45	37[2d]	NR
									Secukinumab 300 mg (SC LD) (100)	7.4 ± 7.5	46.9 ± 12.6	49	33[2d]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
FUTURE 3[33]	✓	✓	✓	✓	✓	✓	✓	CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (137)	6.6 ± 6.9	50.1 ± 12.6	56.9	32.1[2d]	NR
									Secukinumab 150 mg (SC LD) (138)	7.7 ± 8.5	50.1 ± 11.7	55.8	31.8[2d]	NR
									Secukinumab 300 mg (SC LD) (139)	8.3 ± 9.2	49.3 ± 12.9	51.8	31.7[2d]	NR
FUTURE 4[34]	✓	✓	✓	✓	✓	✓	✓	CASPAR criteria; ≥3 swollen joints and ≥3 tender joints;	Placebo (114)	6.9 ± 7.6	48.5 ± 12.12	60.5	23.7[2d]	NR
									Secukinumab 150 mg (no LD) (113)	5.7 ± 7.7	50.4 ± 11.78	54.9	24[2d]	NR
									Secukinumab 150 mg (SC LD) (114)	5.6 ± 7.3	48.3 ± 12.17	58.8	23.7[2d]	NR
FUTURE 5[35]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints;	Placebo (332)	6.6 ± 7.6	49 ± 12.1	51.5	29.5[2d]	NR
									Secukinumab 150 mg (no LD) (222)	6.2 ± 6.1	48.8 ± 11.8	45.9	28.8[2d]	NR
									Secukinumab 150 mg (SC LD) (220)	6.7 ± 7.1	48.4 ± 12.9	49.5	29.5[2d]	NR
									Secukinumab 300 mg (SC LD) (222)	6.7 ± 8.3	48.9 ± 12.8	51.4	30.7[2d]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
EXCEED[36]	✓	✓		✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints; active plaque PsO with ≥1 plaque of ≥2 cm diameter or nail changes consistent with PsO or documented history of plaque PsO	Adalimumab 40 mg Q2W (427)	5.7 ± 7.29	49.5 ± 12.44	46	0	100
									Secukinumab 300 mg Q4W (426)	5.1 ± 7.6	48.5 ± 12.38	51	0	100
MAXIMISE[37]	✓	✓						CASPAR criteria	Placebo (166)	NR ± NR	46.6 ± 11.5	47	0	NR
									Secukinumab 150 mg Q4W (165)	NR ± NR	46.9 ± 11.5	50.9	0	NR
									Secukinumab 300 mg Q4W (167)	NR ± NR	46.2 ± 12.3	53.9	0	NR
Gottlieb 2019[38]	✓			✓			✓	6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (79)	7.3 ± 8	48.1 ± 13.3	55.7	23.7[2d]	NR
									Tildrakizumab 20 mg Q12W (78)	6.6 ± 6.7	47.2 ± 13.4	52.6	24.4[2d]	NR
									Tildrakizumab 100 mg Q12W (77)	6.3 ± 7.2	49.2 ± 11.9	61	23.8[2d]	NR
									Tildrakizumab 200 mg Q12W (79)	6.8 ± 6.3	49.3 ± 11.2	46.8	21.8[2d]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
									Tildrakizumab 200 mg Q4W (78)	6.9 ± 7.3	50.1 ± 13.3	59	22.8[2d]	NR
Opal Beyond[39]	✓		✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender or painful joints; active plaque PsO (diagnosed or confirmed by a dermatologist or a sponsor-approved rheumatologist)	Placebo (131)	9.4 ± 8.1	49 ± 12.6	61	100[2d]	NR
									Tofacitinib 5 mg BID (131)	9.6 ± 7.6	49.5 ± 12.3	49	100[2d]	NR
									Tofacitinib 10 mg BID (132)	9.1 ± 6.8	51.3 ± 10.9	56	100[2d]	NR
Opal Broaden[40]	✓	✓		✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (105)	6.4 ± 6.4	47.7 ± 12.3	53	3[2f]	100[2e]
									Adalimumab 40 mg Q2W (106)	5.3 ± 5.3	47.4 ± 11.3	47	1[2f]	100[2e]
									Tofacitinib 5 mg BID (107)	7.3 ± 8.2	49.4 ± 12.6	53	3[2f]	100[2e]
									Tofacitinib 10 mg BID (104)	5.4 ± 5.8	46.9 ± 12.4	60	4[2f]	100[2e]
SELECT-PsA 1[41]	✓			✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (423)[9]	NR ± NR	NR ± NR	NR	NR	100
									Upadacitinib 15 mg QD (429)[9]	NR ± NR	NR ± NR	NR	NR	100
									Upadacitinib 30 mg QD (423)[9]	NR ± NR	NR ± NR	NR	NR	100

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
									Adalimumab 40 mg Q2W (429) [9]	NR ± NR	NR ± NR	NR	NR	100
SELECT-PsA 2[42]	✓		✓	✓			✓	6 months diagnosis; CASPAR criteria; ≥3 swollen joints and ≥3 tender joints	Placebo (212)	NR ± NR	NR ± NR	NR	100	NR
									Upadacitinib 15 mg QD (211)	NR ± NR	NR ± NR	NR	100	NR
									Upadacitinib 30 mg QD (218)	NR ± NR	NR ± NR	NR	100	NR
Gottlieb 2009[43]	✓			✓			✓	≥6 months diagnosis; ≥3 swollen joints and ≥3 tender joints; either CRP ≥15 mg/L or morning stiffness ≥45 minutes; active plaque PsO, with a qualifying target lesion of ≥2 cm	Placebo (70)	4.9; 2.43-10.71[10]	47.5; 40-55[10]	47	31[2c]	59
									Ustekinumab 90 mg (76)	6.15; 2.78-13.70[10]	50; 42-60.5[10]	41	24[2c]	63
PSUMMIT 1[44]	✓	✓		✓	✓	✓	✓	≥6 months diagnosis; ≥5 swollen joints and ≥5 tender joints; high-sensitivity CRP level of ≥3.0 mg/dL and active or documented history of plaque PsO	Placebo (206)	3.6; 1.0–9.7[10]	47.4 ± 12.29	47.6	0[2c]	NR
									Ustekinumab 45 mg (205)	3.4; 1.2–9.2[10]	47.1 ± 12.64	48.3	0[2c]	NR
									Ustekinumab 90mg (204)	4.9; 1.7–8.3[10]	46.8 ± 11.75	43.1	0[2c]	NR
PSUMMIT 2[45]	✓	✓	✓	✓	✓	✓	✓	≥6 months diagnosis; CASPAR criteria; ≥5 swollen joints and	Placebo (104)	5.5; 2.3-12.2[10]	48, 38.5-56.0[10]	51	59.6[2d]	NR

Study	ACR			PASI	EnthR	DactR	DAE	PsA related inclusion criteria	Interventions and comparator(s) (n)	Duration of PsA in years: (mean±SD)	Age in years (mean±SD)	Female (%)	Prior bDMARD use (%)	Prior csDMARD use (%)
	All	bDMARD-naïve	bDMARD-exposed											
								≥5 tender joints; high-sensitivity CRP level of ≥6.0 mg/dL	Ustekinumab 45 mg (103)	5.3; 2.3-12.2[10]	49; 40.0-56.0[10]	53.4	58.3[2d]	NR
									Ustekinumab 90mg (105)	4.5; 1.7-10.3[10]	48; 41.0-57.0[10]	53.3	55.2[2d]	NR

Abbreviations: ACR, American College of Rheumatology; bDMARD, biologic disease-modifying antirheumatic drug; BID, twice daily; BIW, twice weekly; BSA, body surface area; CASPAR, Classification Criteria for Psoriatic Arthritis; CRP, C reactive protein; csDMARD, conventional synthetic disease-modifying antirheumatic drug; DactR, resolution of dactylitis; DAE, discontinuation due to adverse event; DIP, distal interphalangeal; EnthR, resolution of enthesitis; ESR, erythrocyte sedimentation rate; IV, intravenous; LD, loading dose; LEF, leflunomide mg, milligrams; mg/dL, milligrams per decilitre; mg/L, milligrams per litre; mm/h, millimeters per hour; n, number of patients randomised; NR, not reported; PASI, Psoriasis Area and Severity Index; PGA, Physicians Global Assessment; PsA, psoriatic arthritis; PsO, psoriasis; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks; RF-, rheumatoid factor negative; SC, subcutaneous; SD, standard deviation; SSZ, sulfasalazine; TNF, tumor necrosis factor

[1] Number of patients included in the analysis; 246 patients randomised in total

[2a] Methotrexate exclusively

[2b] Anti-TNF and anti-interleukin-12/23

[2c] Any biologic

[2d] Anti-TNF or anti-TNFα

[2e] Any csDMARD

[2f] Non-TNF inhibitor

[3] Data reported at week 24

[4] Data reported at week 16

[5] GO-DACT was included in the NMA for the outcome *resolution of dactylitis* only

[6] Mean and SD converted from months to years by dividing by 12

[7] Timepoint converted from months to weeks by dividing by 4

[8] Data reported at week 12

[9] Number of patients included in the analysis; 1705 patients randomised in total

[10] Median; IQR

Risk of bias

Random sequence generation

Of the 46 included studies, 31 (67%) reported low risk methods for sequence generation. The remaining 15 studies did not provide sufficient information to allow for assessment, therefore the risk of bias was unclear.

Allocation concealment

Low risk allocation methods were reported in 16 (35%) studies. Central randomisation was the most frequently reported method of concealment (11 of 16 studies). The allocation protocol of the remaining 30 studies was not reported, therefore assessment was unable to be conducted.

Blinding of participants, personnel, and outcome assessment

Forty-four studies were low risk since there was sufficient blinding of participants and personnel (96%). Conversely, two studies (4%) were open-labelled studies and were assigned high risk of bias. The outcome assessment methodologies of nine studies (20%) did not provide enough detail to determine the level of bias. Low risk of bias was declared in the remaining 36 studies (78%) where masking of outcome assessors was clearly defined. Open-label trial designs was reported in one study (2%) therefore, high risk of bias was determined.

Incomplete outcome data

Outcomes were described in sufficient detail in 36 studies (78%). However, four studies (9%) did not adequately report on attritions or exclusion and were assessed to have a high risk of bias. Six studies (13%) reported variation in discontinuation between treatment arms without justification. As a result, there was an unclear level of bias.

Selective reporting

Thirty-eight studies met the criteria for low risk of bias. High risk was assessed in five studies for failing to report on a pre-defined outcome of interest. The remaining three studies were assigned unclear selective reporting due to unclear protocol or unavailability of the primary publication.

Other bias

Seven studies[3,8,11,14,23,25] indicated potential for other sources of bias (high risk), six studies failed to meet the target sample size, one study had discrepancies in baseline characteristics between treatment arms. Ten studies[24,27-29,34,35,37,38,42,46] were assigned unclear risk as it was not possible to determine sources of bias. Of these unclear studies, five studies[29,37,38,42,46] did not have a primary publication, one[24] had heterogeneity in results between publications, two studies[27,28] did not report or obtain the target sample size to prove statistical power, and the remaining studies reported variation in baseline characteristics[34] and attrition rates.[35] All studies were industry funded.

Figure S 1 Risk of Bias graph

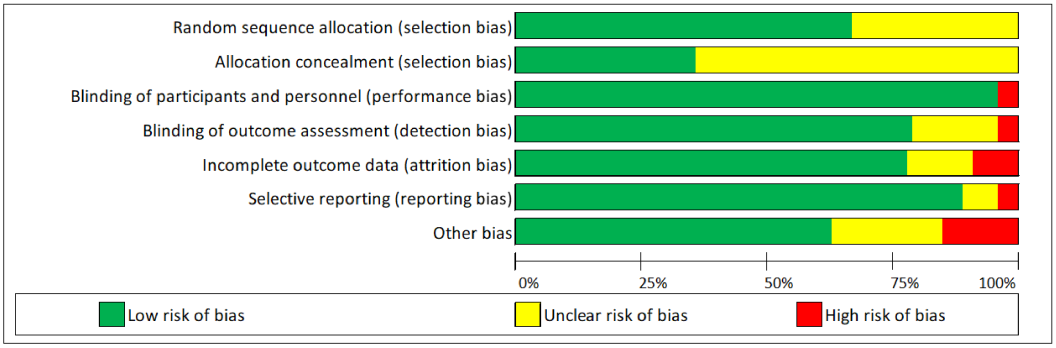
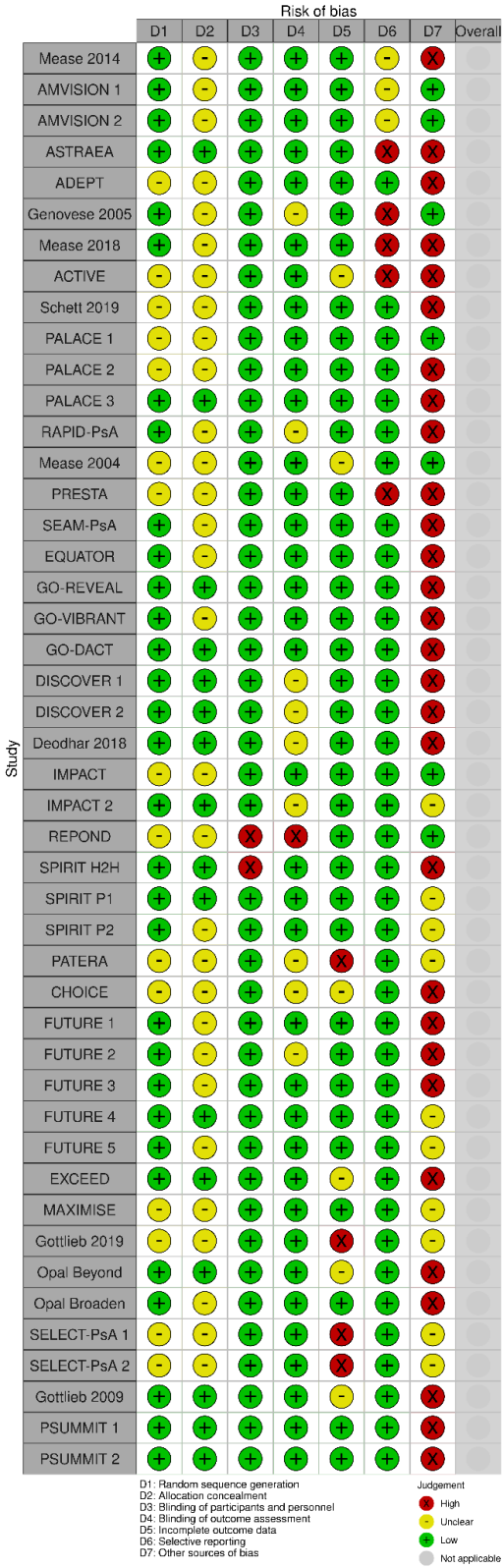


Figure S 2 Concise critical appraisal checklist for included studies



Model fit

The best fitting models varied across outcomes and analyses. ACR response rates for both the all-patient and bDMARD-naïve populations used a random-effect model with placebo adjustment. The ACR bDMARD-experienced population used an unadjusted fixed-effect model. PASI response was modelled using a fixed-effect model with placebo adjustment. Random-effects and fixed-effects models with placebo adjustment provided the best fit for enthesitis and dactylitis analyses, respectively. An unadjusted fixed-effect model provided the best fit for the DAE analysis.

Table S 5 Model fit statistics

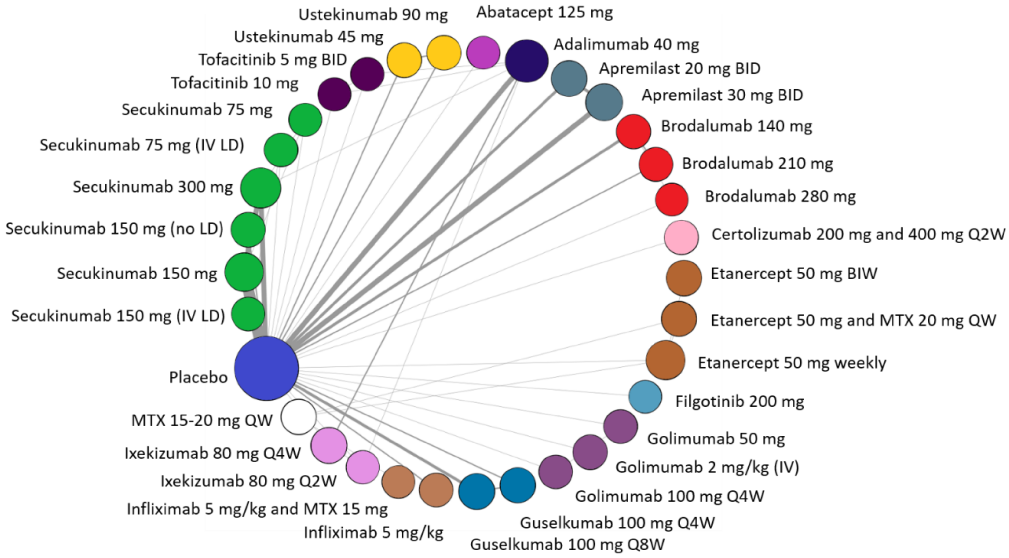
	Model fit statistics	**Unadjusted model**		**Baseline risk adjusted model**	
		Fixed effect	**Random effects**	**Fixed effect**	**Random effects**
ACR – ITT	No. data points	346	*346*	-	346
	Total residual deviance	647.2	*627.3*	-	626.9
	DIC	2225.2	*2219.6*	-	2223.5
	Standard deviation (sd)	NA	*0.110 (0.00936, 0.193)*	-	0.1078 (0.04153, 0.1716)
	Adjustment coefficient (B)	NA	*NA*	-	-0.67 (-1.018, -0.363)
ACR – bDMARD-naïve	No. data points	228	*228*	-	228
	Total residual deviance	359.4	*344.6*	-	345
	DIC	1398.4	*1395.2*	-	1397.8
	Standard deviation (sd)	NA	*0.131 (0.00941, 0.241)*	-	0.125 (0.04843, 0.2176)
	Adjustment coefficient (B)	NA	*NA*	-	-0.764 (-1.312, -0.34)
ACR – bDMARD-experienced	No. data points	*132*	132	132	-
	Total residual deviance	*197.8*	195.8	201.4	-
	DIC	*691.9*	693.3	691.3	-
	Standard deviation (sd)	*NA*	0.138 (0.00782, 0.407)	NA	-
	Adjustment coefficient (B)	*NA*	NA	-1.165 (-2.112, 0.053)	-
PASI	No. data points	*113*	113	113	-
	Total residual deviance	*254.5*	256.2	260.8	-
	DIC	*1026.3*	1028.5	1033.4	-
	Standard deviation (sd)	*NA*	0.142 (0.00495, 0.271)	NA	-
	Adjustment coefficient (B)	*NA*	NA	-0.816 (-1.07, -0.561	-

	Model fit statistics	Unadjusted model		Baseline risk adjusted model	
		Fixed effect	**Random effects**	**Fixed effect**	**Random effects**
Resolution of enthesitis	No. data points	70	*70*	-	70
	Total residual deviance	77.9	*70.1*	-	71.5
	DIC	459.2	*458.8*	-	462.8
	Standard deviation (sd)	NA	*0.195 (0.0165, 0.410)*	-	0.206
	Adjustment coefficient (B)	NA	*NA*	-	-0.534 (-0.964, -0.05)
Resolution of dactylitis	No. data points	*66*	66	66	-
	Total residual deviance	*61.6*	60.9	63.9	-
	DIC	*389.7*	391.4	392.8	-
	Standard deviation (sd)	*NA*	0.102 (0.00476, 0.347)	NA	-
	Adjustment coefficient (B)	*NA*	NA	-0.629 (-0.898, -0.258)	-
DAE	No. data points	*119*	119	-	-
	Total residual deviance	*112.7*	112.8	-	-
	DIC	*560.2*	562.8	-	-
	Standard deviation (sd)	*NA*	0.116 (0.005, 0.433)	-	-
	Adjustment coefficient (B)	*NA*	NA	-	-

Italicised values indicate a better fit between fixed and random effects models; the baseline-risk adjusted model was only run for the better fitting unadjusted model.

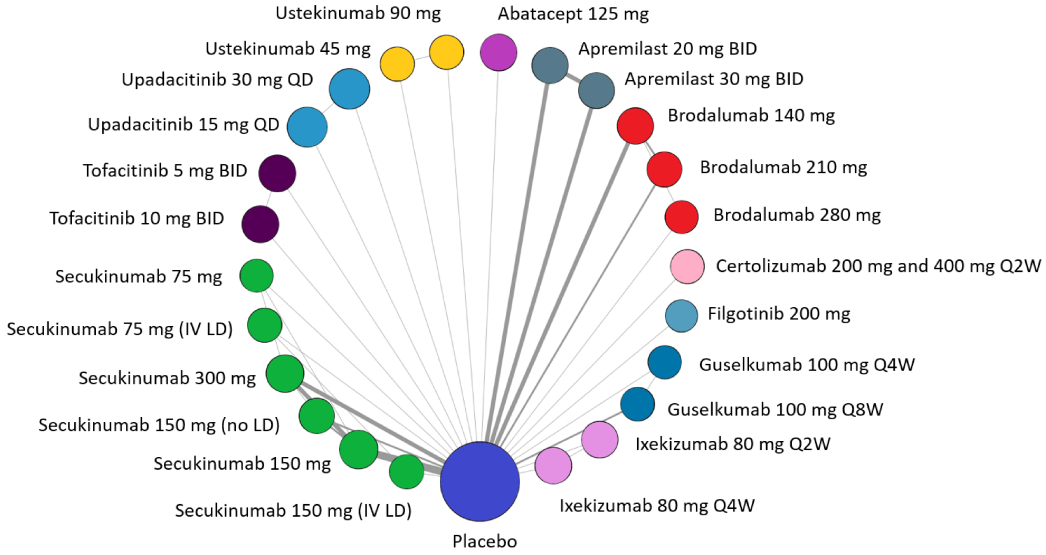
Abbreviations: ACR, American College of Rheumatology; bDMARD, biologic disease-modifying antirheumatic drug; DAE, discontinuation due to adverse event; DIC, Deviance Information Criterion; ITT, intention to treat; PASI, Psoriasis Area and Severity Index; sd, standard deviation.

Figure S 3. Network diagram for ACR response in bDMARD-naïve subgroup



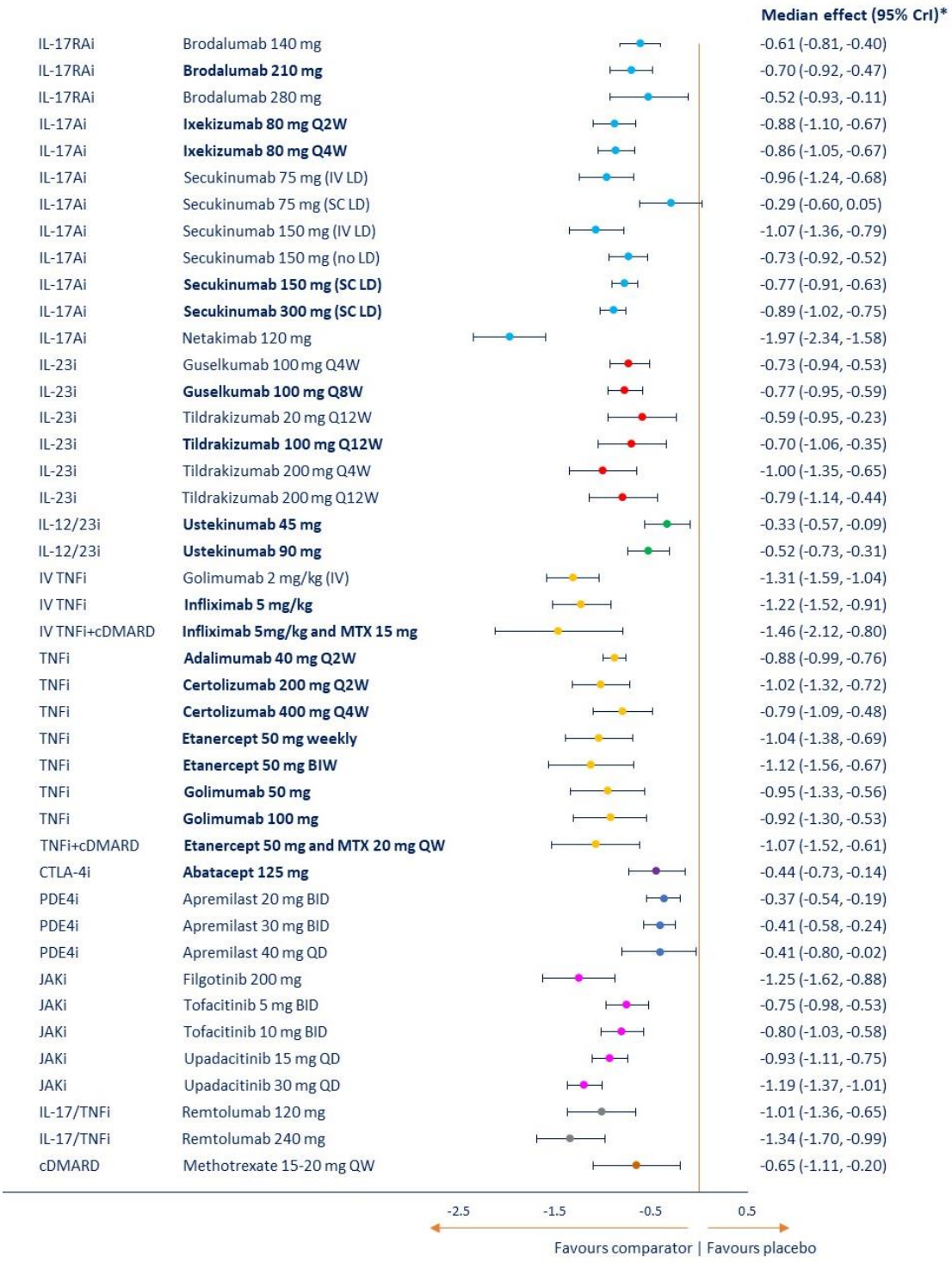
Node size denotes total number of patients on that treatment; line thickness denotes total number of studies informing that comparison.
Abbreviations: BID, twice daily; IV, intravenous; LD, loading dose; MTX, methotrexate; mg, milligram; mg/kg, milligram per kilogram; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks.

Figure S 4. Network diagram for ACR response in bDMARD-experienced subgroup



Node size denotes total number of patients on that treatment; line thickness denotes total number of studies informing that comparison.
Abbreviations: BID, twice daily; IV, intravenous; LD, loading dose; mg, milligram; QD, daily; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks.

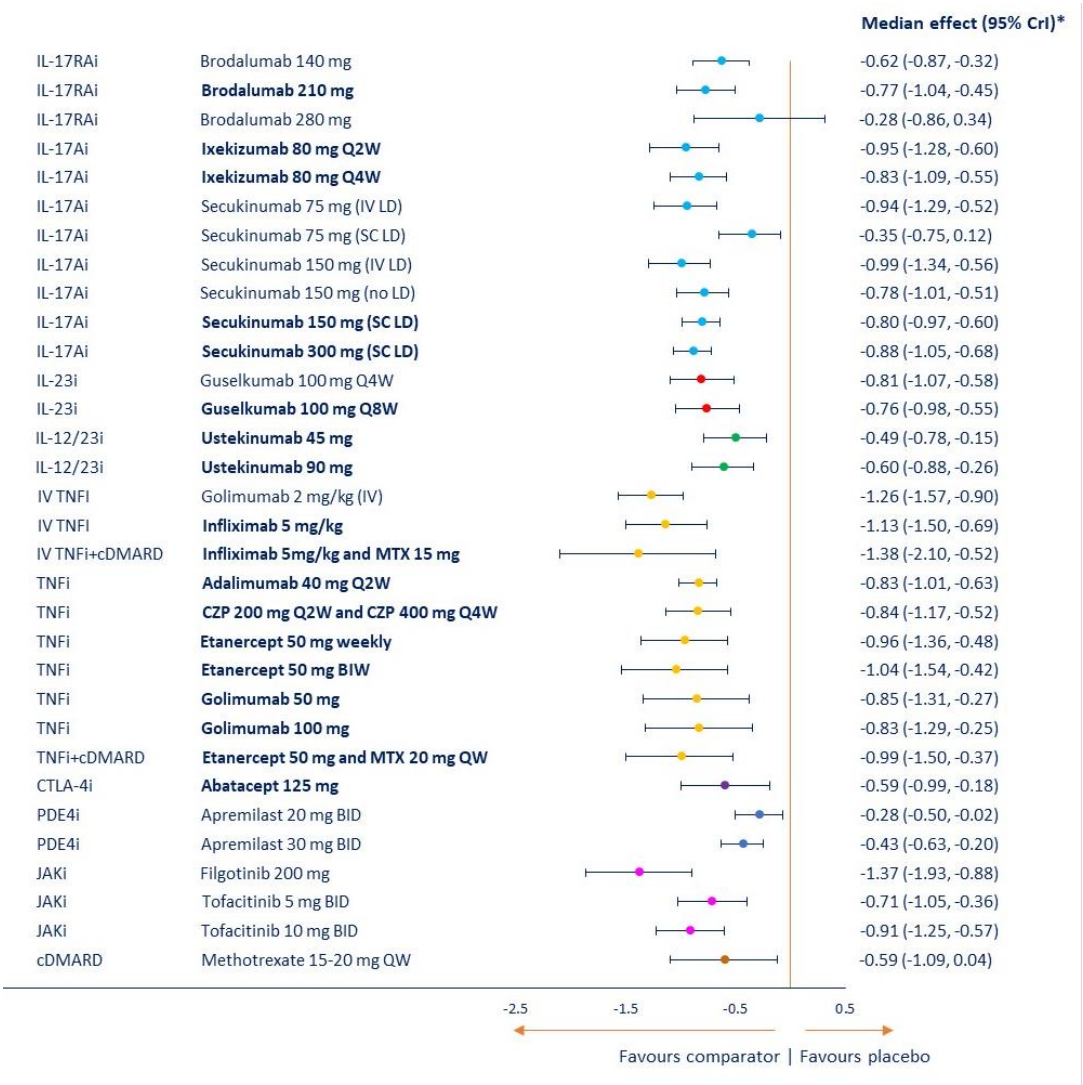
Figure S 5 Forest plot of treatment effects for all interventions versus placebo on ACR response



ACR overall treatment effect was based on a random-effects model with placebo adjustment. Key comparators are shown in bold.

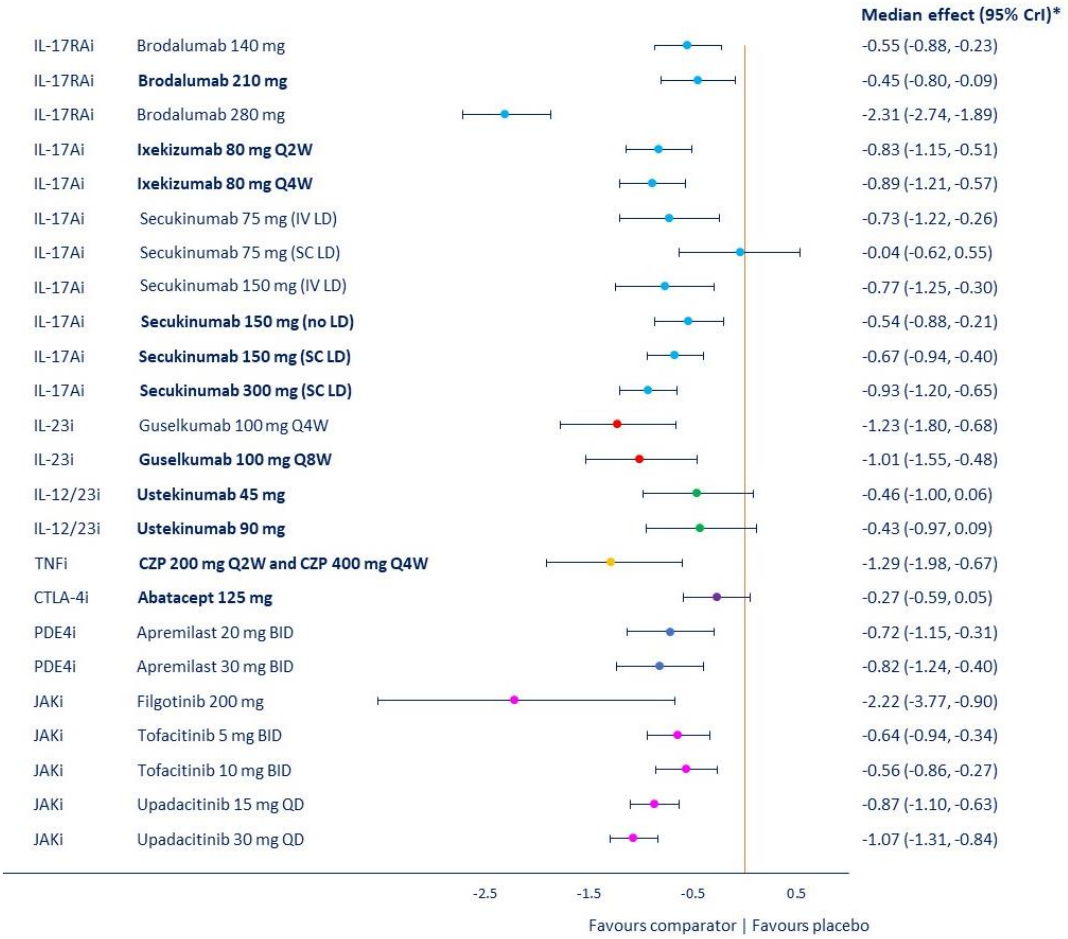
Abbreviations: BID, twice daily; BIW, twice weekly; CrI, credible interval; IV, intravenous; kg, kilogram; LD, loading dose; mg, milligram; MTX, methotrexate; QD, daily; QW, weekly; Q2W, every two weeks; Q4W, every four weeks; Q8W, every eight weeks; Q12W, every twelve weeks; SUCRA, surface under the cumulative ranking curve.

Figure S 6 Forest plot of treatment effects for all interventions versus placebo on ACR response in bDMARD-naïve subgroup



ACR treatment effect was based on a random-effects model with placebo adjustment. Key comparators are shown in bold.
Abbreviations: BID, twice daily; BIW, twice weekly; CrI, credible interval; CZP, certolizumab pegol; IV, intravenous; kg, kilogram; LD, loading dose; mg, milligram; MTX, methotrexate; QD, daily; QW, weekly; Q2W, every two weeks; Q4W, every four weeks; Q8W, every eight weeks; Q12W, every twelve weeks; SUCRA, surface under the cumulative ranking curve.

Figure S 7 Forest plot of treatment effects for all interventions versus placebo on ACR response in bDMARD-experienced subgroup



ACR treatment effect was based on a fixed-effects model with placebo adjustment. Key comparators are shown in bold.
Abbreviations: BID, twice daily; BIW, twice weekly; CrI, credible interval; CZP, certolizumab pegol; IV, intravenous; kg, kilogram; LD, loading dose; mg, milligram; MTX, methotrexate; QD, daily; QW, weekly; Q2W, every two weeks; Q4W, every four weeks; Q8W, every eight weeks; Q12W, every twelve weeks; SUCRA, surface under the cumulative ranking curve.

Table S 6 Median odds ratios (95% credible intervals) for pairwise comparisons of key biologic and selected systemic comparators on achievement of ACR 20

Intervention	Comparator	All patients	bDMARD-naïve	bDMARD-experienced
BRO 210	IXE Q4W	0.77 (0.49, 1.22)	0.91 (0.51, 1.61)	0.49 (0.22, 1.06)
	IXE Q2W	0.74 (0.45, 1.21)	0.75 (0.38, 1.47)	0.53 (0.24, 1.16)
	SEC 150	0.89 (0.59, 1.32)	0.96 (0.6, 1.53)	0.7 (0.33, 1.44)
	SEC 300	0.73 (0.49, 1.09)	0.85 (0.52, 1.36)	**0.46 (0.22, 0.96)**
	GUS Q8W	0.9 (0.56, 1.4)	1.03 (0.59, 1.76)	0.4 (0.14, 1.15)
	TIL Q12W	1 (0.5, 1.94)	-	-
	UST 45	**1.81 (1.09, 3.01)**	1.58 (0.87, 2.85)	0.98 (0.34, 2.82)
	UST 90	1.32 (0.83, 2.11)	1.32 (0.73, 2.41)	1.03 (0.35, 2.97)
	ADA	0.75 (0.51, 1.11)	0.91 (0.56, 1.47)	-
	CZP 200	0.6 (0.32, 1.08)	0.9 (0.45, 1.73)	**0.25 (0.07, 0.82)**
	CZP 400	0.87 (0.47, 1.58)		
	ETA 50	0.58 (0.3, 1.09)	0.74 (0.35, 1.51)	-
	ETA 100	0.51 (0.23, 1.11)	0.65 (0.26, 1.56)	-
	ETA+MTX	0.55 (0.25, 1.22)	0.7 (0.28, 1.7)	-
	GOL 50	0.67 (0.34, 1.3)	0.88 (0.4, 1.95)	-
	GOL 100	0.7 (0.36, 1.36)	0.92 (0.41, 2.03)	-
	INF	**0.43 (0.24, 0.76)**	0.56 (0.28, 1.08)	-
	INF+MTX	**0.29 (0.08, 0.88)**	0.37 (0.09, 1.25)	-
	ABT	1.53 (0.84, 2.77)	1.35 (0.62, 2.88)	1.34 (0.6, 3.02)
	APR 30	**1.59 (1.04, 2.41)**	**1.73 (1.05, 2.85)**	0.55 (0.22, 1.33)
	TOF 5	0.93 (0.55, 1.52)	1.1 (0.54, 2.18)	0.73 (0.34, 1.57)
	UPA 15	0.69 (0.43, 1.08)	-	0.5 (0.25, 1.01)
	Placebo	**3.64 (1.82, 7.65)**	**4.31 (2, 9.61)**	**2.14 (1.16, 3.85)**
IXE Q4W	IXE Q2W	0.96 (0.65, 1.42)	0.82 (0.45, 1.48)	1.09 (0.69, 1.73)
	SEC 150	1.15 (0.81, 1.65)	1.05 (0.66, 1.69)	1.43 (0.73, 2.79)
	SEC 300	0.95 (0.67, 1.35)	0.93 (0.59, 1.48)	0.94 (0.48, 1.85)
	GUS Q8W	1.16 (0.77, 1.74)	1.14 (0.66, 1.91)	0.82 (0.29, 2.21)
	TIL Q12W	1.29 (0.67, 2.42)	-	-
	UST 45	**2.34 (1.45, 3.8)**	1.73 (0.94, 3.21)	2 (0.73, 5.57)
	UST 90	**1.71 (1.11, 2.67)**	1.46 (0.81, 2.69)	2.1 (0.76, 5.91)
	ADA	0.97 (0.72, 1.32)	1 (0.68, 1.49)	-
	CZP 200	0.77 (0.44, 1.36)	0.99 (0.51, 1.89)	0.52 (0.14, 1.59)
	CZP 400	1.12 (0.63, 2)		
	ETA 50	0.75 (0.4, 1.4)	0.81 (0.39, 1.71)	-
	ETA 100	0.66 (0.3, 1.42)	0.71 (0.29, 1.76)	-
	ETA+MTX	0.71 (0.32, 1.56)	0.77 (0.31, 1.92)	-
	GOL 50	0.86 (0.44, 1.7)	0.96 (0.43, 2.25)	-
	GOL 100	0.9 (0.46, 1.78)	1 (0.45, 2.34)	-
	INF	**0.56 (0.31, 0.98)**	0.61 (0.31, 1.23)	-
	INF+MTX	0.37 (0.11, 1.14)	0.4 (0.1, 1.4)	-
	ABT	**1.97 (1.13, 3.45)**	1.48 (0.69, 3.14)	**2.74 (1.31, 5.89)**
	APR 30	**2.06 (1.39, 3.06)**	**1.9 (1.15, 3.14)**	1.12 (0.48, 2.63)
	TOF 5	1.2 (0.75, 1.87)	1.21 (0.62, 2.31)	1.5 (0.75, 3.04)
	UPA 15	0.89 (0.59, 1.33)	-	1.03 (0.55, 1.98)
	Placebo	**4.68 (2.45, 9.87)**	**4.71 (2.23, 10.84)**	**4.39 (2.62, 7.38)**
IXE Q2W	SEC 150	1.19 (0.8, 1.79)	1.28 (0.72, 2.31)	1.31 (0.66, 2.57)

Intervention	Comparator	All patients	bDMARD-naïve	bDMARD-experienced
	SEC 300	0.98 (0.67, 1.47)	1.13 (0.64, 2.03)	0.86 (0.43, 1.69)
	GUS Q8W	1.2 (0.77, 1.89)	1.38 (0.73, 2.56)	0.76 (0.27, 2.02)
	TIL Q12W	1.34 (0.68, 2.6)	-	-
	UST 45	**2.43 (1.46, 4.1)**	**2.11 (1.05, 4.25)**	1.83 (0.67, 5.1)
	UST 90	**1.77 (1.12, 2.89)**	1.76 (0.89, 3.6)	1.92 (0.69, 5.36)
	ADA	1.01 (0.7, 1.47)	1.21 (0.71, 2.13)	-
	CZP 200	0.8 (0.44, 1.46)	1.2 (0.58, 2.48)	0.47 (0.13, 1.46)
	CZP 400	1.17 (0.64, 2.12)		
	ETA 50	0.78 (0.41, 1.5)	0.98 (0.44, 2.26)	-
	ETA 100	0.69 (0.31, 1.52)	0.86 (0.33, 2.31)	-
	ETA+MTX	0.74 (0.33, 1.67)	0.93 (0.35, 2.53)	-
	GOL 50	0.89 (0.45, 1.82)	1.17 (0.48, 2.99)	-
	GOL 100	0.94 (0.47, 1.9)	1.22 (0.5, 3.11)	-
	INF	0.58 (0.32, 1.05)	0.74 (0.34, 1.67)	-
	INF+MTX	0.39 (0.11, 1.2)	0.49 (0.12, 1.82)	-
	ABT	**2.05 (1.15, 3.73)**	1.79 (0.78, 4.09)	**2.51 (1.21, 5.37)**
	APR 30	**2.14 (1.39, 3.32)**	**2.3 (1.27, 4.28)**	1.03 (0.44, 2.39)
	TOF 5	1.24 (0.76, 2.03)	1.47 (0.69, 3.09)	1.37 (0.68, 2.78)
	UPA 15	0.93 (0.59, 1.45)	-	0.94 (0.5, 1.8)
	Placebo	**4.87 (2.48, 10.47)**	**5.72 (2.54, 14.19)**	**4.02 (2.4, 6.75)**
SEC 150	SEC 300	0.82 (0.65, 1.05)	0.88 (0.66, 1.17)	0.66 (0.43, 1)
	GUS Q8W	1.01 (0.7, 1.43)	1.08 (0.7, 1.63)	0.58 (0.21, 1.5)
	TIL Q12W	1.12 (0.6, 2.07)	-	-
	UST 45	**2.04 (1.34, 3.1)**	1.65 (1, 2.69)	1.4 (0.52, 3.78)
	UST 90	**1.49 (1.03, 2.16)**	1.38 (0.85, 2.27)	1.47 (0.55, 4)
	ADA	0.85 (0.65, 1.11)	0.95 (0.68, 1.33)	-
	CZP 200	0.67 (0.39, 1.14)	0.94 (0.52, 1.64)	0.36 (0.1, 1.09)
	CZP 400	0.98 (0.57, 1.66)		
	ETA 50	0.66 (0.36, 1.15)	0.77 (0.4, 1.46)	-
	ETA 100	0.58 (0.27, 1.19)	0.68 (0.29, 1.54)	-
	ETA+MTX	0.62 (0.29, 1.3)	0.73 (0.31, 1.69)	-
	GOL 50	0.75 (0.4, 1.38)	0.92 (0.45, 1.91)	-
	GOL 100	0.79 (0.43, 1.46)	0.95 (0.46, 1.98)	-
	INF	**0.49 (0.29, 0.8)**	0.58 (0.32, 1.04)	-
	INF+MTX	**0.33 (0.09, 0.96)**	0.38 (0.1, 1.25)	-
	ABT	**1.72 (1.02, 2.89)**	1.4 (0.71, 2.78)	1.92 (0.97, 3.92)
	APR 30	**1.8 (1.3, 2.46)**	**1.8 (1.25, 2.61)**	0.78 (0.35, 1.76)
	TOF 5	1.04 (0.67, 1.58)	1.15 (0.63, 2.08)	1.05 (0.55, 2.02)
	UPA 15	0.78 (0.54, 1.1)	-	0.72 (0.41, 1.28)
	Placebo	**4.09 (2.17, 8.22)**	**4.48 (2.25, 9.44)**	**3.08 (1.98, 4.79)**
SEC 300	GUS Q8W	1.22 (0.85, 1.73)	1.22 (0.79, 1.85)	0.88 (0.32, 2.3)
	TIL Q12W	1.36 (0.73, 2.49)	-	-
	UST 45	**2.47 (1.62, 3.76)**	**1.86 (1.12, 3.09)**	2.13 (0.8, 5.78)
	UST 90	**1.8 (1.24, 2.62)**	1.56 (0.96, 2.59)	2.23 (0.84, 6.11)
	ADA	1.03 (0.8, 1.31)	1.07 (0.8, 1.46)	-
	CZP 200	0.82 (0.48, 1.37)	1.06 (0.59, 1.87)	0.55 (0.16, 1.66)
	CZP 400	1.18 (0.69, 2.01)		
	ETA 50	0.8 (0.44, 1.4)	0.87 (0.45, 1.66)	-

Intervention	Comparator	All patients	bDMARD-naïve	bDMARD-experienced
	ETA 100	0.7 (0.33, 1.44)	0.77 (0.33, 1.75)	-
	ETA+MTX	0.75 (0.35, 1.58)	0.83 (0.35, 1.92)	-
	GOL 50	0.91 (0.49, 1.68)	1.04 (0.51, 2.19)	-
	GOL 100	0.95 (0.51, 1.77)	1.08 (0.52, 2.28)	-
	INF	**0.59 (0.35, 0.97)**	0.66 (0.36, 1.19)	-
	INF+MTX	0.39 (0.11, 1.16)	0.43 (0.12, 1.42)	-
	ABT	**2.09 (1.23, 3.48)**	1.59 (0.8, 3.17)	**2.93 (1.47, 5.97)**
	APR 30	**2.18 (1.57, 2.98)**	**2.04 (1.4, 2.98)**	1.2 (0.53, 2.68)
	TOF 5	1.26 (0.82, 1.9)	1.3 (0.71, 2.34)	1.6 (0.83, 3.08)
	UPA 15	0.94 (0.65, 1.33)	-	1.1 (0.62, 1.97)
	Placebo	**4.95 (2.64, 9.96)**	**5.08 (2.55, 10.79)**	**4.69 (2.98, 7.32)**
GUS Q8W	TIL Q12W	1.11 (0.59, 2.07)	-	-
	UST 45	**2.02 (1.25, 3.32)**	1.53 (0.86, 2.76)	2.43 (0.72, 8.48)
	UST 90	1.48 (0.95, 2.33)	1.28 (0.74, 2.33)	2.55 (0.75, 8.85)
	ADA	0.84 (0.6, 1.18)	0.88 (0.58, 1.38)	-
	CZP 200	0.67 (0.38, 1.17)	0.87 (0.48, 1.59)	0.63 (0.15, 2.44)
	CZP 400	0.97 (0.55, 1.7)		
	ETA 50	0.65 (0.34, 1.23)	0.72 (0.34, 1.49)	-
	ETA 100	0.57 (0.26, 1.25)	0.63 (0.26, 1.55)	-
	ETA+MTX	0.62 (0.28, 1.38)	0.68 (0.27, 1.7)	-
	GOL 50	0.74 (0.37, 1.51)	0.85 (0.37, 2.01)	-
	GOL 100	0.78 (0.39, 1.58)	0.89 (0.39, 2.09)	-
	INF	**0.48 (0.26, 0.87)**	0.54 (0.27, 1.09)	-
	INF+MTX	**0.32 (0.09, 1)**	0.36 (0.09, 1.24)	-
	ABT	1.7 (0.99, 2.95)	1.3 (0.65, 2.65)	**3.33 (1.21, 9.56)**
	APR 30	**1.78 (1.2, 2.67)**	**1.68 (1.07, 2.67)**	1.36 (0.46, 4.18)
	TOF 5	1.03 (0.66, 1.61)	1.07 (0.57, 2.01)	1.82 (0.68, 4.97)
	UPA 15	0.77 (0.52, 1.15)	-	1.25 (0.49, 3.29)
	Placebo	**4.02 (2.15, 8.64)**	**4.11 (2.1, 9.49)**	**5.31 (2.27, 13.06)**
TIL Q12W	UST 45	1.82 (0.91, 3.7)	-	-
	UST 90	1.33 (0.68, 2.64)	-	-
	ADA	0.76 (0.42, 1.39)	-	-
	CZP 200	0.6 (0.28, 1.26)	-	-
	CZP 400	0.87 (0.41, 1.84)	-	-
	ETA 50	0.58 (0.26, 1.33)	-	-
	ETA 100	0.51 (0.2, 1.32)	-	-
	ETA+MTX	0.55 (0.21, 1.43)	-	-
	GOL 50	0.67 (0.28, 1.64)	-	-
	GOL 100	0.7 (0.29, 1.72)	-	-
	INF	**0.43 (0.2, 0.96)**	-	-
	INF+MTX	0.29 (0.07, 1.01)	-	-
	ABT	1.54 (0.74, 3.21)	-	-
	APR 30	1.6 (0.85, 3.08)	-	-
	TOF 5	0.93 (0.48, 1.8)	-	-
	UPA 15	0.69 (0.37, 1.3)	-	-
	Placebo	**3.63 (1.64, 9.04)**	-	-
UST 45	UST 90	0.73 (0.49, 1.1)	0.84 (0.52, 1.38)	1.05 (0.46, 2.4)
	ADA	**0.41 (0.27, 0.64)**	**0.58 (0.35, 0.97)**	-

Intervention	Comparator	All patients	bDMARD-naïve	bDMARD-experienced
	CZP 200	**0.33 (0.17, 0.62)**	0.57 (0.28, 1.13)	**0.26 (0.06, 0.96)**
	CZP 400	**0.48 (0.25, 0.9)**		
	ETA 50	**0.32 (0.17, 0.61)**	**0.47 (0.22, 0.98)**	-
	ETA 100	**0.28 (0.13, 0.62)**	0.41 (0.17, 1.01)	-
	ETA+MTX	**0.3 (0.13, 0.68)**	0.44 (0.18, 1.1)	-
	GOL 50	**0.37 (0.19, 0.72)**	0.56 (0.26, 1.24)	-
	GOL 100	**0.39 (0.2, 0.75)**	0.58 (0.26, 1.29)	-
	INF	**0.24 (0.13, 0.42)**	**0.35 (0.18, 0.69)**	-
	INF+MTX	**0.16 (0.04, 0.5)**	**0.23 (0.06, 0.81)**	-
	ABT	0.84 (0.45, 1.57)	0.85 (0.39, 1.89)	1.37 (0.49, 3.87)
	APR 30	0.88 (0.57, 1.36)	1.1 (0.65, 1.85)	0.56 (0.18, 1.7)
	TOF 5	**0.51 (0.29, 0.87)**	0.7 (0.34, 1.42)	0.75 (0.28, 2.02)
	UPA 15	**0.38 (0.23, 0.62)**	-	0.52 (0.2, 1.33)
	Placebo	2.02 (0.98, 4.23)	**2.74 (1.24, 6.16)**	2.2 (0.9, 5.27)
UST 90	ADA	**0.57 (0.39, 0.83)**	0.69 (0.42, 1.13)	-
	CZP 200	**0.45 (0.25, 0.82)**	0.68 (0.33, 1.33)	**0.25 (0.06, 0.94)**
	CZP 400	0.66 (0.36, 1.19)		
	ETA 50	**0.44 (0.23, 0.81)**	0.56 (0.27, 1.14)	-
	ETA 100	**0.39 (0.18, 0.82)**	0.49 (0.2, 1.18)	-
	ETA+MTX	**0.42 (0.19, 0.9)**	0.53 (0.21, 1.29)	-
	GOL 50	**0.5 (0.27, 0.94)**	0.66 (0.31, 1.45)	-
	GOL 100	**0.53 (0.28, 0.99)**	0.69 (0.32, 1.51)	-
	INF	**0.33 (0.19, 0.56)**	**0.42 (0.22, 0.8)**	-
	INF+MTX	**0.22 (0.06, 0.66)**	**0.28 (0.07, 0.94)**	-
	ABT	1.16 (0.64, 2.07)	1.01 (0.46, 2.22)	1.31 (0.46, 3.75)
	APR 30	1.21 (0.81, 1.78)	1.31 (0.77, 2.16)	0.53 (0.17, 1.63)
	TOF 5	0.7 (0.42, 1.15)	0.83 (0.4, 1.68)	0.71 (0.26, 1.93)
	UPA 15	**0.52 (0.33, 0.81)**	-	0.49 (0.19, 1.27)
	Placebo	**2.76 (1.38, 5.65)**	**3.26 (1.47, 7.24)**	2.09 (0.85, 5.03)
ADA	CZP 200	0.8 (0.47, 1.33)	0.99 (0.55, 1.74)	-
	CZP 400	1.15 (0.68, 1.94)		-
	ETA 50	0.77 (0.42, 1.38)	0.81 (0.42, 1.55)	-
	ETA 100	0.68 (0.32, 1.42)	0.72 (0.31, 1.64)	-
	ETA+MTX	0.73 (0.34, 1.55)	0.77 (0.33, 1.79)	-
	GOL 50	0.89 (0.47, 1.67)	0.97 (0.47, 2.03)	-
	GOL 100	0.93 (0.49, 1.76)	1.01 (0.49, 2.11)	-
	INF	**0.57 (0.34, 0.96)**	0.62 (0.34, 1.11)	-
	INF+MTX	0.38 (0.11, 1.14)	0.4 (0.11, 1.32)	-
	ABT	**2.03 (1.23, 3.36)**	1.48 (0.74, 2.95)	-
	APR 30	**2.12 (1.54, 2.91)**	**1.9 (1.29, 2.77)**	-
	TOF 5	1.23 (0.83, 1.79)	1.21 (0.68, 2.12)	-
	UPA 15	0.92 (0.67, 1.25)	-	-
	Placebo	**4.8 (2.62, 9.83)**	**4.73 (2.37, 10.04)**	-
CZP 200*	CZP 400	1.45 (0.83, 2.55)	-	-
	ETA 50	0.97 (0.46, 2.06)	0.82 (0.36, 1.89)	-
	ETA 100	0.85 (0.35, 2.06)	0.72 (0.27, 1.94)	-
	ETA+MTX	0.92 (0.37, 2.25)	0.78 (0.29, 2.11)	-
	GOL 50	1.11 (0.5, 2.51)	0.98 (0.39, 2.55)	-

Intervention	Comparator	All patients	bDMARD-naïve	bDMARD-experienced
	GOL 100	1.17 (0.53, 2.63)	1.01 (0.41, 2.64)	-
	INF	0.72 (0.36, 1.47)	0.62 (0.28, 1.39)	-
	INF+MTX	0.48 (0.13, 1.6)	0.41 (0.1, 1.51)	-
	ABT	**2.56 (1.31, 5.08)**	1.49 (0.67, 3.32)	**5.31 (1.68, 19.39)**
	APR 30	**2.67 (1.54, 4.71)**	**1.92 (1.07, 3.53)**	2.17 (0.64, 8.5)
	TOF 5	1.55 (0.85, 2.83)	1.22 (0.59, 2.56)	2.9 (0.94, 10.2)
	UPA 15	1.15 (0.66, 2.04)	-	1.99 (0.69, 6.78)
	Placebo	**6.08 (2.87, 14.14)**	**4.73 (2.17, 11.88)**	**8.45 (3.08, 27.65)**
CZP 400*	ETA 50	0.67 (0.31, 1.41)	-	-
	ETA 100	0.59 (0.24, 1.41)	-	-
	ETA+MTX	0.64 (0.26, 1.54)	-	-
	GOL 50	0.77 (0.35, 1.71)	-	-
	GOL 100	0.81 (0.37, 1.79)	-	-
	INF	0.5 (0.25, 1)	-	-
	INF+MTX	0.33 (0.09, 1.1)	-	-
	ABT	1.76 (0.9, 3.46)	-	-
	APR 30	**1.84 (1.05, 3.22)**	-	-
	TOF 5	1.07 (0.58, 1.94)	-	-
	UPA 15	0.79 (0.45, 1.4)	-	-
	Placebo	**4.17 (1.96, 9.7)**	-	-
ETA 50	ETA 100	0.88 (0.56, 1.37)	0.88 (0.53, 1.46)	-
	ETA+MTX	0.95 (0.59, 1.51)	0.95 (0.56, 1.59)	-
	GOL 50	1.14 (0.54, 2.46)	1.19 (0.53, 2.8)	-
	GOL 100	1.2 (0.57, 2.58)	1.24 (0.55, 2.89)	-
	INF	0.74 (0.38, 1.47)	0.76 (0.36, 1.58)	-
	INF+MTX	0.5 (0.17, 1.24)	0.5 (0.16, 1.33)	-
	ABT	**2.63 (1.25, 5.51)**	1.82 (0.73, 4.53)	-
	APR 30	**2.73 (1.53, 4.99)**	**2.34 (1.2, 4.58)**	-
	TOF 5	1.59 (0.8, 3.15)	1.49 (0.64, 3.47)	-
	UPA 15	1.19 (0.63, 2.26)	-	-
	Placebo	**6.31 (2.76, 14.41)**	**5.91 (2.34, 14.14)**	-
ETA 100	ETA+MTX	1.08 (0.57, 2.07)	1.08 (0.52, 2.24)	-
	GOL 50	1.3 (0.55, 3.21)	1.35 (0.53, 3.71)	-
	GOL 100	1.36 (0.58, 3.36)	1.41 (0.54, 3.85)	-
	INF	0.84 (0.38, 1.94)	0.86 (0.35, 2.15)	-
	INF+MTX	0.56 (0.18, 1.58)	0.56 (0.17, 1.71)	-
	ABT	**3 (1.25, 7.19)**	2.07 (0.73, 5.92)	-
	APR 30	**3.11 (1.5, 6.67)**	**2.66 (1.14, 6.23)**	-
	TOF 5	1.82 (0.79, 4.13)	1.7 (0.62, 4.57)	-
	UPA 15	1.35 (0.62, 2.99)	-	-
	Placebo	**7.18 (2.81, 18.52)**	**6.71 (2.34, 18.45)**	-
ETA+MTX	GOL 50	1.2 (0.5, 3)	1.25 (0.48, 3.43)	-
	GOL 100	1.26 (0.53, 3.17)	1.31 (0.5, 3.58)	-
	INF	0.78 (0.34, 1.82)	0.79 (0.33, 1.98)	-
	INF+MTX	0.52 (0.19, 1.32)	0.52 (0.18, 1.4)	-
	ABT	**2.78 (1.14, 6.77)**	1.92 (0.66, 5.52)	-
	APR 30	**2.88 (1.37, 6.27)**	**2.47 (1.05, 5.83)**	-
	TOF 5	1.68 (0.73, 3.86)	1.57 (0.57, 4.26)	-

Intervention	Comparator	All patients	bDMARD-naïve	bDMARD-experienced
	UPA 15	1.25 (0.57, 2.8)	-	-
	Placebo	**6.65 (2.56, 17.35)**	**6.22 (2.15, 17.16)**	-
GOL 50	GOL 100	1.05 (0.6, 1.82)	1.04 (0.57, 1.89)	-
	INF	0.65 (0.33, 1.25)	0.63 (0.3, 1.29)	-
	INF+MTX	0.43 (0.11, 1.42)	0.42 (0.1, 1.47)	-
	ABT	**2.3 (1.02, 5.05)**	1.53 (0.55, 4.07)	-
	APR 30	**2.4 (1.29, 4.42)**	1.97 (0.92, 4.09)	-
	TOF 5	1.39 (0.65, 2.9)	1.26 (0.47, 3.17)	-
	UPA 15	1.04 (0.51, 2.06)	-	-
	Placebo	**5.57 (2.27, 12.52)**	**5.01 (1.78, 11.98)**	-
GOL 100	INF	0.62 (0.31, 1.19)	0.61 (0.29, 1.24)	-
	INF+MTX	0.41 (0.11, 1.36)	0.4 (0.1, 1.42)	-
	ABT	2.19 (0.98, 4.8)	1.47 (0.53, 3.99)	-
	APR 30	**2.28 (1.23, 4.21)**	1.89 (0.89, 3.96)	-
	TOF 5	1.33 (0.62, 2.76)	1.21 (0.45, 3.06)	-
	UPA 15	0.99 (0.49, 1.96)	-	-
	Placebo	**5.3 (2.17, 11.93)**	**4.82 (1.71, 11.65)**	-
INF	INF+MTX	0.67 (0.18, 2.1)	0.66 (0.17, 2.22)	-
	ABT	**3.55 (1.75, 7.17)**	2.41 (0.99, 5.86)	-
	APR 30	**3.7 (2.22, 6.2)**	**3.09 (1.69, 5.69)**	-
	TOF 5	**2.15 (1.13, 4.05)**	1.97 (0.86, 4.5)	-
	UPA 15	1.6 (0.89, 2.88)	-	-
	Placebo	**8.58 (3.85, 18.15)**	**7.87 (3.16, 17.4)**	-
INF+MTX	ABT	**5.3 (1.62, 19.6)**	3.67 (0.94, 15.81)	-
	APR 30	**5.51 (1.85, 19.11)**	**4.7 (1.42, 17.76)**	-
	TOF 5	**3.21 (1.01, 11.38)**	3 (0.8, 12.18)	-
	UPA 15	2.38 (0.78, 8.45)	-	-
	Placebo	**12.74 (3.67, 48.99)**	**11.89 (3.04, 48.78)**	-
ABT	APR 30	1.04 (0.6, 1.81)	1.29 (0.64, 2.59)	**0.41 (0.17, 0.96)**
	TOF 5	0.61 (0.33, 1.08)	0.82 (0.36, 1.87)	0.55 (0.26, 1.13)
	UPA 15	**0.45 (0.26, 0.78)**	-	**0.38 (0.19, 0.72)**
	Placebo	**2.36 (1.14, 5.51)**	**3.19 (1.33, 8.45)**	1.6 (0.91, 2.72)
APR 30	TOF 5	**0.58 (0.36, 0.91)**	0.64 (0.34, 1.18)	1.15 (0.5, 2.67)
	UPA 15	**0.43 (0.29, 0.64)**	-	0.79 (0.36, 1.71)
	Placebo	**2.29 (1.18, 4.6)**	**2.49 (1.22, 5.33)**	**3.36 (1.7, 6.66)**
TOF 5	UPA 15	0.74 (0.48, 1.17)	-	0.69 (0.37, 1.27)
	Placebo	**3.89 (2.01, 8.67)**	**3.87 (1.73, 9.88)**	**2.93 (1.79, 4.77)**
UPA 15	Placebo	**5.23 (2.78, 11.13)**	-	**4.25 (2.9, 6.26)**

Bolded values indicate a statistically significant difference.

*CZP 200 effects presented for the bDMARD-naïve and bDMARD-exposed subgroups reflect a pooled effect for CZP 200 and 400 doses.

ACR, American College of Rheumatology; ABT, abatacept; ADA, adalimumab 40 mg Q2W; bDMARD, biologic disease modifying anti-rheumatic drug; BID, twice daily; BRO 210, brodalumab 210 mg; CZP 200, certolizumab pegol 200 mg; CZP 400, certolizumab 400 mg; ETA 50, etanercept 50 mg per week; ETA 100, etanercept 100 mg per week; GOL 50, golimumab 50 mg; GOL 100, golimumab 100 mg; GUS, guselkumab 100 mg; INF, infliximab 5 mg/kg; IXE, ixekizumab 80 mg; MTX, methotrexate; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks; SEC 150, secukinumab 150 mg; SEC 300, secukinumab 300 mg; TIL, tildrakizumab 100 mg; TOF 5, tofacitinib 5 mg BID; UPA 15, upadacitinib 15 mg QD; UST 45, ustekinumab 45 mg; UST 90, ustekinumab 90 mg

Table S 7 Probabilities of ACR response by bDMARD exposure subgroup for all comparators

Drug class	Intervention	Probability of achieving outcome, median (95% CrI)								
		ITT			bDMARD-naïve			bDMARD-exposed		
		ACR 20	ACR 50	ACR 70	ACR 20	ACR 50	ACR 70	ACR 20	ACR 50	ACR 70
IL-17RAi	Brodalumab 140 mg	45.1% (35, 55.9)	21.9% (15, 30.8)	8.3% (15, 30.8)	46.9% (35.2, 58.7)	23.4% (15.2, 33.6)	10% (15.2, 33.6)	36.2% (19.4, 56.6)	17.3% (7.3, 33.7)	6.7% (7.3, 33.7)
	Brodalumab 210 mg	48.6% (37.6, 59.7)	24.6% (16.7, 34.3)	9.8% (16.7, 34.3)	53.1% (40.4, 65.3)	28.5% (18.7, 40)	13% (18.7, 40)	32.3% (15.9, 53.1)	14.7% (5.6, 30.6)	5.4% (5.6, 30.6)
	Brodalumab 280 mg	41.4% (26, 59.1)	19.3% (9.8, 33.7)	7% (9.8, 33.7)	34.1% (15.5, 58.1)	14.5% (4.8, 32.9)	5.4% (4.8, 32.9)	92% (79.6, 97.7)	79.2% (59.3, 92)	60% (59.3, 92)
IL-17Ai	Ixekizumab 80 mg Q2W	56% (44.5, 66.6)	30.9% (21.5, 41.3)	13.4% (21.5, 41.3)	60.2% (45.1, 74.2)	34.9% (22.1, 50.1)	17.3% (22.1, 50.1)	47.2% (28.1, 67)	25.5% (12.1, 44.1)	11.2% (12.1, 44.1)
	Ixekizumab 80 mg Q4W	55.1% (44.5, 64.8)	30% (21.5, 39.3)	12.9% (21.5, 39.3)	55.5% (43.1, 67.7)	30.6% (20.6, 42.5)	14.4% (20.6, 42.5)	49.4% (29.9, 69)	27.3% (13.2, 46.4)	12.2% (13.2, 46.4)
	Netakimab 120 mg	89% (80, 95)	71.8% (57.5, 84)	48.7% (57.5, 84)						
	Secukinumab 75 mg (IV LD)	59% (45.7, 71.2)	33.6% (22.4, 46.4)	15.1% (22.4, 46.4)	59.8% (44, 73.4) [b]	34.5% (21.3, 49.1) [b]	17% (21.3, 49.1) [b]	43.2% (21.5, 67.4) [b]	22.4% (8.4, 44.6) [b]	9.4% (8.4, 44.6) [b]
	Secukinumab 75 mg (SC LD)	32.8% (20.6, 46.7)	13.6% (7.1, 23.2)	4.4% (7.1, 23.2)	36.5% (21.3, 53.5) [b]	16.1% (7.4, 28.8) [b]	6.1% (7.4, 28.8) [b]	19.5% (5.7, 43.6) [b]	7.4% (1.5, 22.8) [b]	2.2% (1.5, 22.8) [b]
	Secukinumab 150 mg (IV LD)	63.3% (50.2, 74.9)	37.8% (25.9, 50.8)	17.9% (25.9, 50.8)	61.5% (45.8, 74.9) [b]	36.1% (22.6, 51) [b]	18.1% (22.6, 51) [b]	44.7% (22.6, 68.8) [b]	23.5% (9, 46.1) [b]	10% (9, 46.1) [b]
	Secukinumab 150 mg (no LD)	49.7% (39.2, 60.2)	25.5% (17.7, 34.7)	10.3% (17.7, 34.7)	53.4% (41.8, 64.6)	28.7% (19.6, 39.2)	13.2% (19.6, 39.2)	35.9% (19, 56.3)	17.1% (7.1, 33.4)	6.6% (7.1, 33.4)
	Secukinumab 150 mg (SC LD)	51.5% (42.7, 60.4)	27% (20.2, 35)	11.1% (20.2, 35)	54.2% (45.4, 63.3) [a]	29.4% (22.2, 38) [a]	13.6% (22.2, 38) [a]	40.7% (23.7, 59.7) [a]	20.5% (9.6, 36.6) [a]	8.3% (9.6, 36.6) [a]
	Secukinumab 300 mg (SC LD)	56.3% (47.4, 64.8)	31.2% (23.7, 39.4)	13.6% (23.7, 39.4)	57.2% (48.2, 66.3) [a]	32.1% (24.4, 41.1) [a]	15.4% (24.4, 41.1) [a]	51% (32.2, 69.4) [a]	28.6% (14.7, 46.9) [a]	13.1% (14.7, 46.9) [a]
IL-23i	Guselkumab 100 mg Q4W	49.9% (38.1, 60.3)	25.7% (17, 34.9)	10.4% (17, 34.9)	54.7% (42.3, 66.8) [a]	29.9% (20, 41.6) [a]	13.9% (20, 41.6) [a]	54.2% (28.8, 78.2) [b]	31.4% (12.5, 57.6) [b]	14.9% (12.5, 57.6) [b]
	Guselkumab 100 mg Q8W	51.6% (40.5, 61)	27.1% (18.6, 35.5)	11.1% (18.6, 35.5)	52.6% (41, 63.4) [a]	28.1% (19.1, 38.1) [a]	12.8% (19.1, 38.1) [a]	63% (36.2, 84.8) [b]	39.8% (17.3, 67) [b]	20.7% (17.3, 67) [b]
	Tildrakizumab 20 mg Q12W	44.4% (28.9, 59.9)	21.4% (11.3, 34.5)	8.1% (11.3, 34.5)						
	Tildrakizumab 100 mg Q12W	48.9% (32.9, 64.3)	24.9% (13.7, 38.8)	9.9% (13.7, 38.8)						
	Tildrakizumab 200 mg Q12W	52.3% (36.1, 67.3)	27.7% (15.7, 42)	11.5% (15.7, 42)						

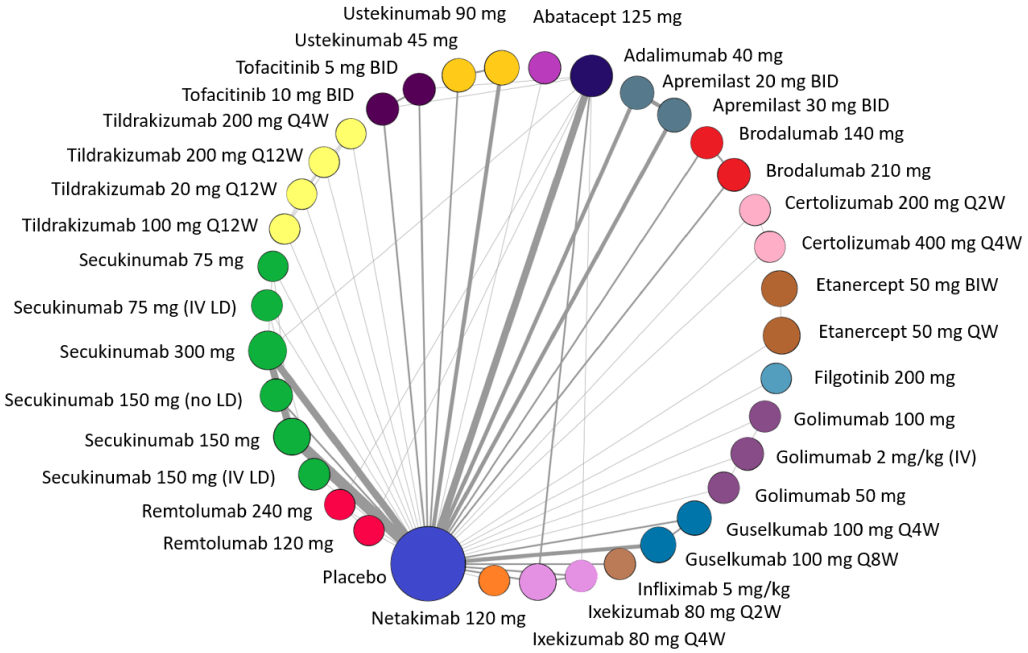
Drug class	Intervention	Probability of achieving outcome, median (95% CrI)								
		ITT			bDMARD-naïve			bDMARD-exposed		
		ACR 20	ACR 50	ACR 70	ACR 20	ACR 50	ACR 70	ACR 20	ACR 50	ACR 70
	Tildrakizumab 200 mg Q4W	60.7% (44.6, 74.4)	35.3% (21.5, 50.2)	16.2% (21.5, 50.2)						
IL-12/23i	Ustekinumab 45 mg	34.2% (24.6, 45.9)	14.5% (9.1, 22.5)	4.8% (9.1, 22.5)	41.8% (29.7, 55.1)	19.6% (11.9, 30.2)	7.9% (11.9, 30.2)	32.9% (13.6, 59)	15.1% (4.6, 36)	5.6% (4.6, 36)
	Ustekinumab 90 mg	41.6% (32, 52.7)	19.4% (13.2, 28)	7.1% (13.2, 28)	46% (33.5, 59)	22.8% (14.1, 33.8)	9.6% (14.1, 33.8)	31.9% (13, 57.9)	14.5% (4.3, 34.9)	5.3% (4.3, 34.9)
TNFi	Adalimumab 40 mg Q2W	55.8% (46.5, 63.6)	30.7% (23, 38.1)	13.3% (23, 38.1)	55.5% (46.3, 64.8)	30.6% (22.9, 39.4)	14.4% (22.9, 39.4)			
	Certolizumab 200 mg Q2W	61.3% (47.1, 73.8)	35.8% (23.5, 49.4)	16.6% (23.5, 49.4)	55.8% (41, 70.3)	30.9% (19.1, 45.4)	14.6% (19.1, 45.4)	65.3% (36.4, 88.1)	42.2% (17.5, 72.3)	22.5% (17.5, 72.3)
	Certolizumab 400 mg Q4W	52.2% (37.9, 65.7)	27.6% (16.9, 40.3)	11.5% (16.9, 40.3)						
	Etanercept 50 mg weekly	61.8% (47.2, 75.6)	36.3% (23.6, 51.7)	16.9% (23.6, 51.7)	60.3% (42.8, 75.1)	35% (20.3, 51.3)	17.3% (20.3, 51.3)			
	Etanercept 50 mg BIW	64.8% (46.7, 80.4)	39.4% (23.2, 58.1)	19% (23.2, 58.1)	63.3% (40.8, 80.4)	38% (19, 58.3)	19.5% (19, 58.3)			
	Etanercept 50 mg and Methotrexate 20 mg QW	63% (44.4, 79.4)	37.5% (21.5, 56.7)	17.7% (21.5, 56.7)	61.6% (38.7, 79.3)	36.2% (17.5, 56.7)	18.2% (17.5, 56.7)			
	Golimumab 50 mg	58.4% (43.3, 73.9)	33% (20.6, 49.6)	14.7% (20.6, 49.6)	55.9% (35.3, 73.3)	30.9% (15.3, 48.9)	14.6% (15.3, 48.9)			
	Golimumab 100 mg	57.2% (42, 72.9)	32% (19.7, 48.3)	14.1% (19.7, 48.3)	55% (34.6, 72.5)	30.1% (14.8, 48)	14.1% (14.8, 48)			
IV TNFi	Golimumab 2 mg/kg (IV)	71.8% (60.1, 81.6)	47.1% (34.7, 59.9)	24.8% (34.7, 59.9)	71.4% (58, 82.2)	46.7% (32.9, 60.9)	26.2% (32.9, 60.9)			
	Infliximab 5 mg/kg	68.4% (56.7, 79.9)	43.2% (31.5, 57.5)	21.8% (31.5, 57.5)	66.6% (51.6, 79.3)	41.4% (27.2, 56.8)	22% (27.2, 56.8)			
	Infliximab 5mg/kg and Methotrexate 15 mg	76.5% (52, 92.1)	52.9% (27.4, 77.7)	29.6% (27.4, 77.7)	75.3% (45, 92.2)	51.5% (22, 77.9)	30.2% (22, 77.9)			
CTLA-4i	Abatacept 125 mg	38.4% (25.7, 51.6)	17.2% (9.6, 27.1)	6% (9.6, 27.1)	45.9% (29.1, 63.8)	22.7% (11.5, 38.4)	9.6% (11.5, 38.4)	26.4% (12.6, 45.1)	11.1% (4.1, 23.9)	3.8% (4.1, 23.9)
IL-17/TNFi	Remtolumab 120 mg	61% (44.6, 74.7)	35.5% (21.5, 50.7)	16.4% (21.5, 50.7)						
	Remtolumab 240 mg	73% (58, 84.2)	48.5% (32.6, 63.8)	25.9% (32.6, 63.8)						
PDE4i	Apremilast 20 mg BID	35.5% (27.4, 45.5)	15.3% (10.5, 22.2)	5.2% (10.5, 22.2)	33.8% (24.5, 44.2)	14.3% (9, 21.4)	5.3% (9, 21.4)	42.8% (22.6, 65.5)	22.1% (9, 42.5)	9.2% (9, 42.5)

Drug class	Intervention	Probability of achieving outcome, median (95% CrI)								
		ITT			bDMARD-naïve			bDMARD-exposed		
		ACR 20	ACR 50	ACR 70	ACR 20	ACR 50	ACR 70	ACR 20	ACR 50	ACR 70
	Apremilast 30 mg BID	37.2% (29, 46.9)	16.4% (11.4, 23.3)	5.6% (11.4, 23.3)	39.6% (30.4, 49.6)	18.1% (12.3, 25.5)	7.1% (12.3, 25.5)	46.5% (25.4, 69.1)	25% (10.5, 46.4)	10.9% (10.5, 46.4)
	Apremilast 40 mg QD	37.3% (23, 54)	16.5% (8.2, 29.1)	5.7% (8.2, 29.1)						
JAKi	Filgotinib 200 mg	70% (53.4, 82.4)	44.9% (28.6, 61)	23.1% (28.6, 61)	75.2% (55.2, 90.1)	51.3% (30.3, 73.8)	30% (30.3, 73.8)	90.8% (47.4, 99.8)	77% (25.6, 99)	57.2% (25.6, 99)
	Tofacitinib 5 mg BID	50.7% (38.6, 61.6)	26.4% (17.3, 36.1)	10.7% (17.3, 36.1)	50.9% (35.5, 66.5)	26.6% (15.4, 41.2)	11.9% (15.4, 41.2)	39.5% (22.2, 59.3)	19.7% (8.8, 36.3)	7.9% (8.8, 36.3)
	Tofacitinib 10 mg BID	52.8% (40.5, 63.6)	28.1% (18.7, 38.1)	11.7% (18.7, 38.1)	58.6% (43.3, 73.2)	33.4% (20.7, 48.8)	16.2% (20.7, 48.8)	36.7% (20.1, 56.2)	17.7% (7.7, 33.4)	6.9% (7.7, 33.4)
	Upadacitinib 15 mg QD	58% (47.1, 67.2)	32.7% (23.4, 41.9)	14.5% (23.4, 41.9)				48.6% (30.9, 66.7)	26.7% (13.8, 43.8)	11.9% (13.8, 43.8)
	Upadacitinib 30 mg QD	67.6% (57.2, 75.8)	42.3% (31.9, 51.9)	21.1% (31.9, 51.9)				56.7% (38.5, 73.6)	33.8% (18.9, 51.8)	16.4% (18.9, 51.8)
cDMARD	Methotrexate 15-20 mg QW	46.8% (29, 65.7)	23.2% (11.4, 40.3)	9% (11.4, 40.3)	45.5% (24.5, 65.8)	22.4% (9, 40.5)	9.4% (9, 40.5)			
	Placebo	20.7% (9.5, 37.3)	7.1% (2.5, 16.5)	1.9% (2.5, 16.5)	20.9% (10.2, 36.4)	7.2% (2.7, 16)	2.2% (2.7, 16)	18.3% (9.6, 30.7)	6.8% (2.9, 13.8)	2% (2.9, 13.8)

ACR, American College of Rheumatology; bDMARD, biologic disease modifying anti-rheumatic drug; BID, twice daily; cs, conventional synthetic; i, inhibitor; IL, interleukin; IV, intravenous; JAK, janus kinas; LD, loading dose; PDE4, Phosphodiesterase-4; QD, once daily; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks; SC, subcutaneous; TNF, tumour necrosis factor

[a] Based on a combination of studies reporting outcomes at week 12 or 16 and week 24; [b] Based on studies reporting outcomes at week 24 only

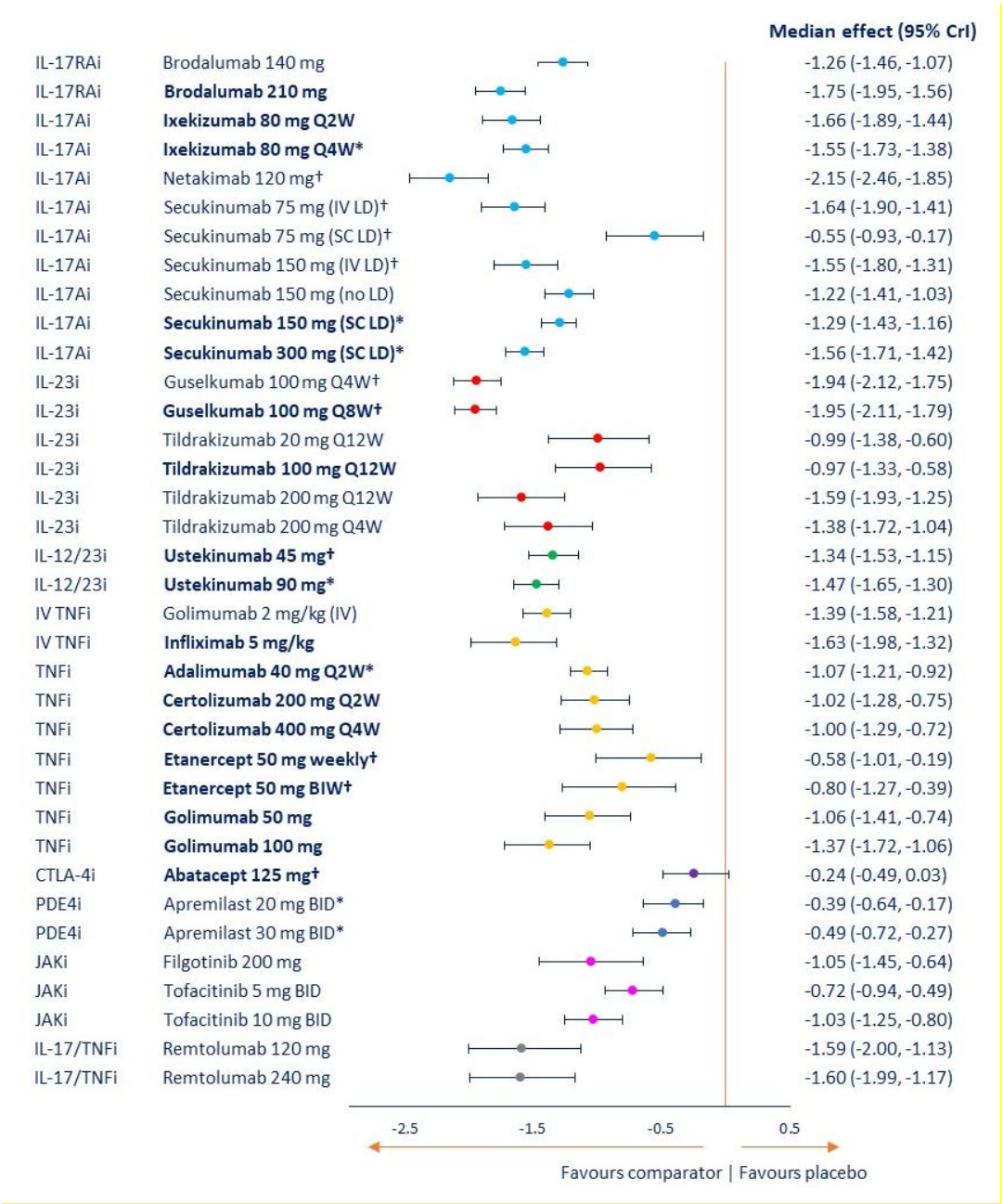
Figure S 8. Network diagram for PASI response



Node size denotes total number of patients on that treatment; line thickness denotes total number of studies informing that comparison.

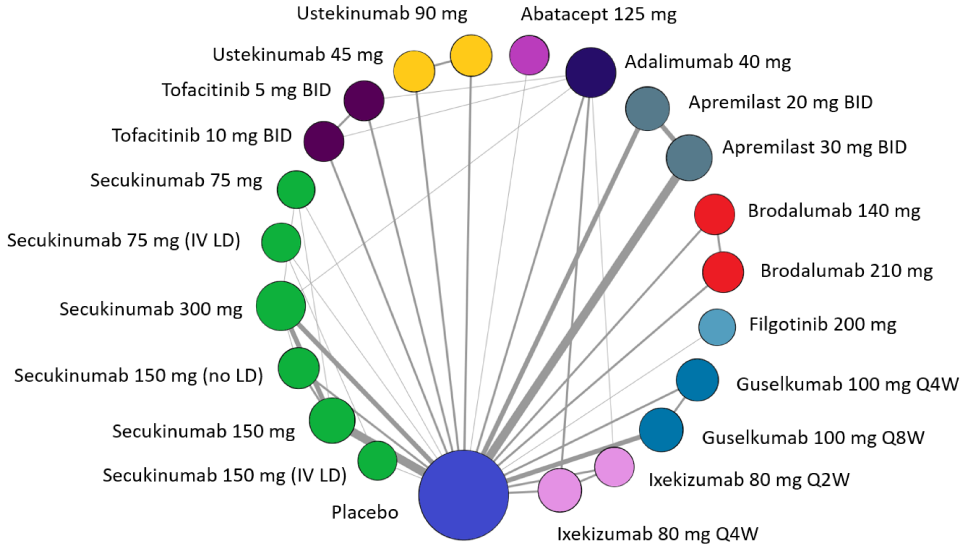
Abbreviations: BID, twice daily; IV, intravenous; LD, loading dose; mg, milligram; mg/kg, milligram per kilogram; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks.

Figure S 9 Forest plot of treatment effects for all interventions versus placebo on PASI response



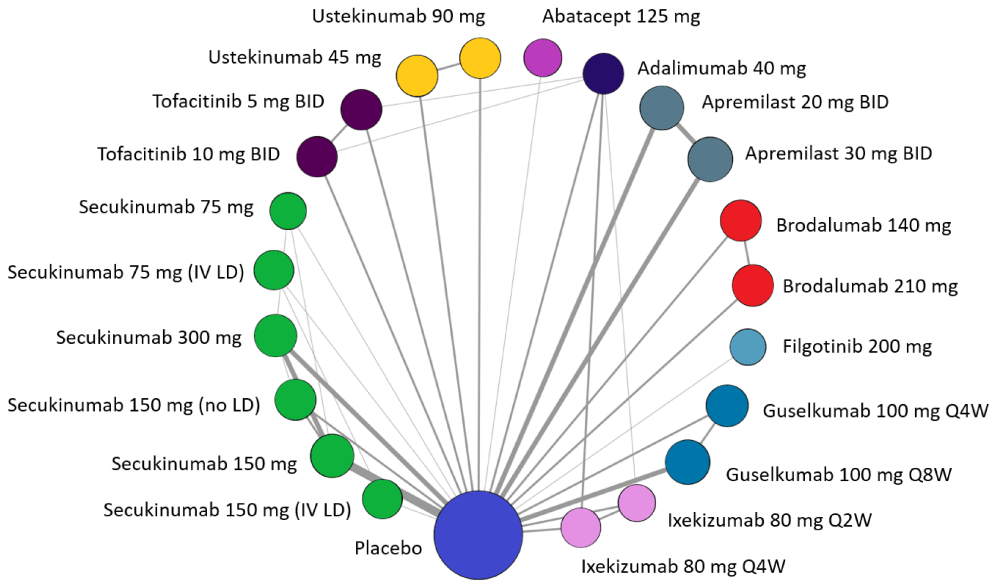
PASI treatment effect was based on a fixed-effects model with placebo adjustment. Median treatment effects and 95% credible intervals plotted on the probit scale. Key comparators are shown in bold.
Abbreviations: BID, twice daily; CrI, credible interval; IV, intravenous; kg, kilogram; LD, loading dose; mg, milligram; Q2W, every two weeks; Q4W, every four weeks; Q12W, every twelve weeks.
* Based on a combination of studies reporting outcomes at week 12 or 16 and week 24; †Based on studies reporting outcomes at week 24 only

Figure S 10 Network diagram for resolution of enthesitis



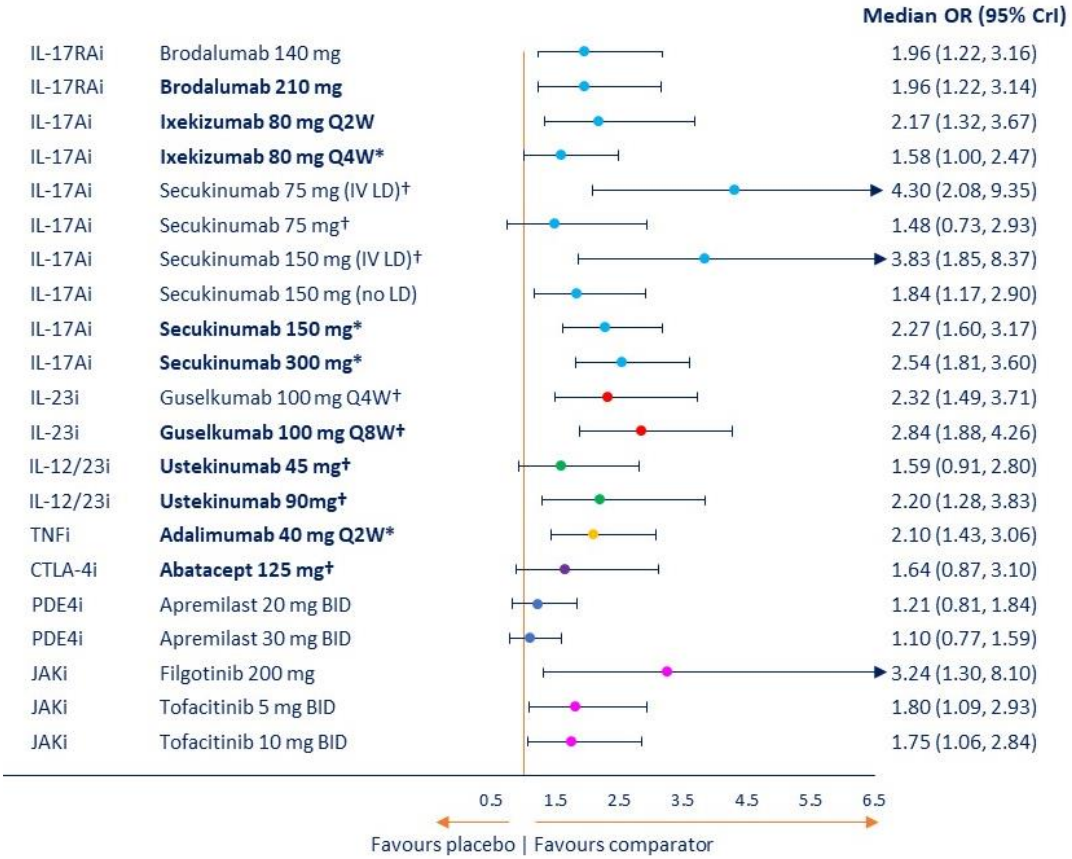
Node size denotes total number of patients on that treatment; line thickness denotes total number of studies informing that comparison.
Abbreviations: BID, twice daily; IV, intravenous; LD, loading dose; mg, milligram; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks.

Figure S 11 Network diagram for resolution of dactylitis



Node size denotes total number of patients on that treatment; line thickness denotes total number of studies informing that comparison.
Abbreviations: BID, twice daily; IV, intravenous; LD, loading dose; mg, milligram; MTX, methotrexate; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks.

Figure S 12 Forest plot for all interventions versus placebo on resolution of enthesitis

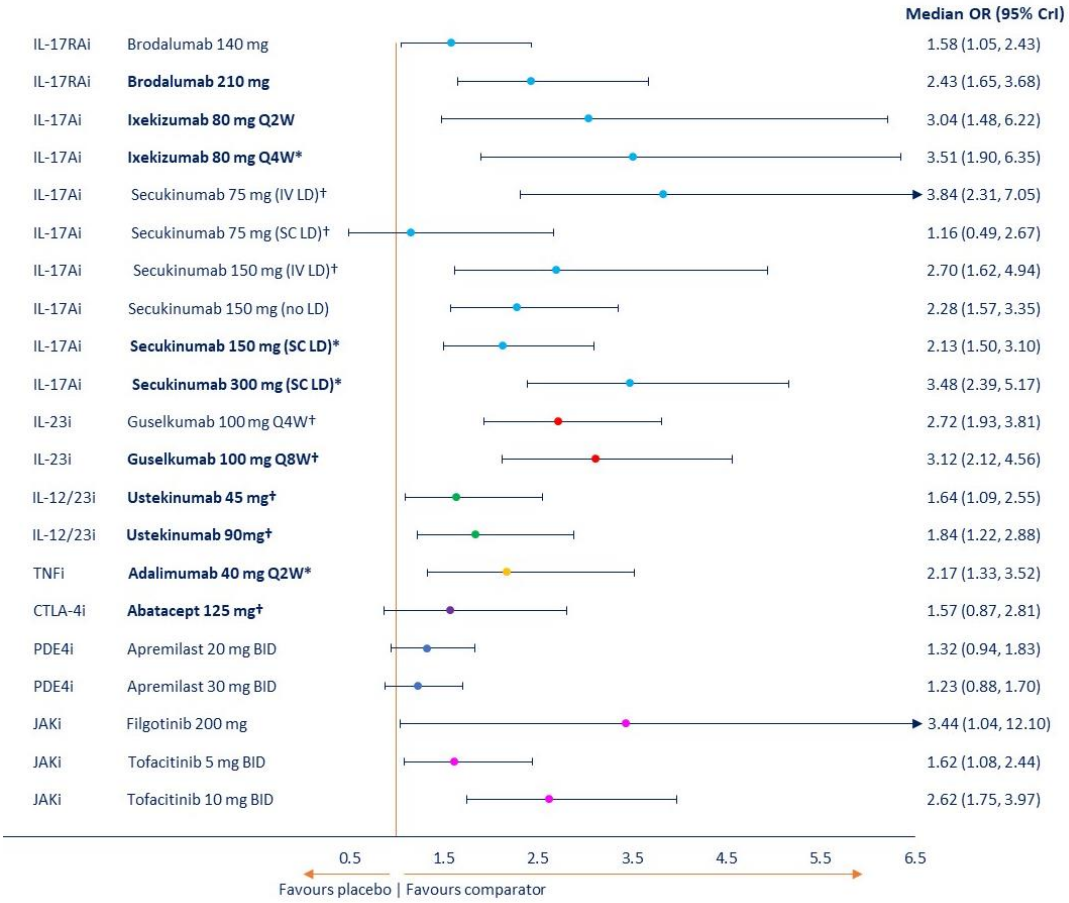


Treatment effect on resolution of enthesitis was based on a random-effects model without placebo adjustment. Key comparators are shown in bold.

Abbreviations: BID, twice daily; CrI, credible interval; LD, loading dose; mg, milligram; OR, odds ratio; Q2W, every two weeks; Q4W, every four weeks

* Based on a combination of studies reporting outcomes at week 12 or 16 and week 24; †Based on studies reporting outcomes at week 24 only

Figure S 13 Forest plot of treatment effects for all interventions versus placebo on resolution of dactylitis



Treatment effect on resolution of dactylitis was based on a fixed-effects model without placebo adjustment. Key comparators are shown in bold.

Abbreviations: BID, twice daily; CrI, credible interval; LD, loading dose; mg, milligram; RR, risk ratio; Q2W, every two weeks; Q4W, every four weeks

* Based on a combination of studies reporting outcomes at week 12 or 16 and week 24; †Based on studies reporting outcomes at week 24 only

Table S 8 Probabilities of response by outcome for all comparators

Drug class	Intervention	Percent achieving outcome, median (95% CrI)			
		PASI 75	**PASI 90**	**Resolution of enthesitis**	**Resolution of dactylitis**
IL-17RAi	Brodalumab 140 mg	51.8% (43, 60.2)	33.8% (26, 41.8)	38.1% (24.9, 52.8)	42.9% (28.7, 61.2)
	Brodalumab 210 mg	70.3% (62.3, 77.4)	52.7% (44, 61.3)	38% (24.9, 52.6)	53.6% (38.5, 70.6)
IL-17Ai	Ixekizumab 80 mg Q2W	67% (57.8, 75.5)	49% (39.4, 58.9)	40.5% (26.7, 56.3)	59.8% (36.3, 77.4)
	Ixekizumab 80 mg Q4W [a]	63.1% (55.4, 70.3)	44.9% (37.1, 52.7)	33.1% (21.5, 46.8)	63.2% (41.7, 78.2)
	Netakimab 120 mg [b]	82.4% (72.9, 89.4)	67.9% (55.8, 78.4)		
	Secukinumab 75 mg (IV LD) [b]	66.6% (56.8, 75.8)	48.5% (38.5, 59.2)	57% (38.5, 76.3)	64.4% (48.9, 81.6)
	Secukinumab 75 mg (SC LD) [b]	25.3% (14.3, 39.1)	12.9% (6.3, 22.9)	31.7% (17.2, 50.1)	35.7% (17.1, 60.3)
	Secukinumab 150 mg (IV LD) [b]	63% (53, 72.6)	44.7% (34.8, 55.5)	54.2% (35.8, 74.3)	56% (40.3, 75.6)
	Secukinumab 150 mg (no LD)	50.2% (41.1, 58.4)	32.3% (24.5, 40)	36.6% (23.9, 50.4)	52.4% (36.3, 68.4)
	Secukinumab 150 mg (SC LD) [a]	53.2% (45.7, 59.5)	35% (28.3, 41.1)	41.6% (29.6, 53.7)	50.5% (35.8, 67.2)
	Secukinumab 300 mg (SC LD) [a]	63.7% (56.3, 69.5)	45.4% (37.9, 51.9)	44.4% (32, 56.7)	62.4% (47.4, 77.1)
IL-23i	Guselkumab 100 mg Q4W [b]	77.1% (69.4, 82)	60.9% (51.8, 67.3)	47.3% (32.7, 59.8)	57% (40, 70.8)
	Guselkumab 100 mg Q8W [b]	76.6% (68.5, 82.2)	60.3% (50.7, 67.5)	42.3% (28.1, 56.3)	60.4% (42.4, 73.8)
	Tildrakizumab 20 mg Q12W	41.2% (26, 56.9)	24.6% (13.4, 38.6)		
	Tildrakizumab 100 mg Q12W	40.2% (25.5, 55.1)	23.7% (13, 36.8)		
	Tildrakizumab 200 mg Q12W	64.7% (50.3, 76.5)	46.5% (32.4, 60.1)		
	Tildrakizumab 200 mg Q4W	56.7% (42.1, 69.6)	38.4% (25.3, 51.9)		
IL-12/23i	Ustekinumab 45 mg [b]	60% (52.8, 67.8)	41.6% (34.6, 49.9)	32.9% (21.1, 49.9)	43.9% (29.5, 62.2)
	Ustekinumab 90 mg	54.8% (46.6, 63) [a]	36.5% (29.1, 44.8) [a]	40.4% (27.2, 57.9) [b]	46.8% (32.0, 65.0) [b]
TNFi	Adalimumab 40 mg Q2W [a]	44.2% (36.9, 50.7)	27% (21.2, 32.8)	39.7% (27.6, 52.6)	51.3% (33.3, 68.1)
	Certolizumab 200 mg Q2W	42.2% (31, 53)	25.4% (16.8, 34.8)		
	Certolizumab 400 mg Q4W	41.5% (30, 53.3)	24.8% (16.1, 35.2)		
	Etanercept 50 mg weekly [b]	26.1% (15.4, 42.2)	13.5% (6.8, 25.4)		
	Etanercept 50 mg BIW [b]	33.9% (20.1, 52.5)	18.9% (9.6, 34.3)		
	Golimumab 50 mg	43.5% (32.1, 58.4)	26.5% (17.6, 40)		
	Golimumab 100 mg	55.7% (44, 69.6)	37.4% (26.9, 52)		
IV TNFi	Golimumab 2 mg/kg (IV)	57.1% (47.9, 64.7)	38.7% (30.3, 46.5)		
	Infliximab 5 mg/kg	65.8% (54.7, 78)	47.7% (36.4, 62)		
CTLA-4i	Abatacept 125 mg [b]	16.3% (10.4, 23.9)	7.4% (4.2, 12)	33.9% (20, 51.8)	43.1% (25.2, 62.3)
PDE4i	Apremilast 20 mg BID [a]	20.4% (14.8, 28.8)	9.8% (6.6, 15.3)	27.4% (18.2, 39.9)	39.3% (24.4, 53.8)
	Apremilast 30 mg BID [a]	23.2% (17.1, 31.8)	11.6% (7.8, 17.5)	25.7% (17, 36.6)	37.6% (23.1, 52)
JAKi	Filgotinib 200 mg	43.6% (27.7, 59.7)	26.5% (14.5, 41.4)	50.3% (27.2, 73)	62.8% (29.8, 86.3)
	Tofacitinib 5 mg BID	30.9% (22.7, 39.6)	16.8% (11.2, 23.4)	36% (23.3, 50.9)	43.8% (28.4, 60.9)
	Tofacitinib 10 mg BID	42.4% (32.9, 51.8)	25.6% (18.2, 33.8)	35.3% (22.7, 50.1)	55.8% (39.2, 71.6)
IL-17/TNFi	Remtolumab 120 mg	64.6% (45.6, 78.5)	46.4% (28.3, 62.7)		

	Remtolumab 240 mg	65.1% (47, 78.1)	46.9% (29.5, 62.3)		
	Placebo[1]	9.7% (3.4, 22)	3.9% (1.1, 10.8)	23% (12.7, 37.8)	31.1% (11, 62.3)

BID, twice daily; i, inhibitor; IL, interleukin; IV, intraventous; JAK, janus kinas; LD, loading dose; PASI, Psoriasis Area and Severity Index; PDE4, Phosphodiesterase-4; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks; SC, subcutaneous; TNF, tumour necrosis factor

[a] Based on a combination of studies reporting outcomes at week 12 or 16 and week 24; [b] Based on studies reporting outcomes at week 24 only

Table S 9 Median odds ratios (95% credible intervals) for pairwise comparisons of key biologic and selected systemic comparators from NMA of PASI response and resolution of enthesitis and dactylitis

Intervention	Comparator	PASI 90	Resolution of enthesitis	Resolution of dactylitis
BRO 210	IXE Q4W	1.37 (0.94, 2.03)	1.24 (0.66, 2.37)	0.69 (0.34, 1.48)
	IXE Q2W	1.16 (0.74, 1.81)	0.9 (0.45, 1.76)	0.8 (0.36, 1.88)
	SEC 150	**2.08 (1.48, 2.96)**	0.86 (0.49, 1.53)	1.14 (0.71, 1.85)
	SEC 300	1.34 (0.96, 1.93)	0.77 (0.43, 1.36)	0.7 (0.43, 1.15)
	GUS Q8W	0.72 (0.5, 1.08)	0.69 (0.37, 1.28)	0.89 (0.55, 1.53)
	TIL Q12W	**3.59 (1.85, 7.59)**	-	-
	UST 45	**1.93 (1.29, 2.91)**	1.23 (0.6, 2.52)	1.48 (0.88, 2.49)
	UST 90	**1.56 (1.06, 2.3)**	0.89 (0.43, 1.8)	1.32 (0.78, 2.23)
	ADA	**3.01 (2.12, 4.37)**	0.93 (0.52, 1.7)	1.12 (0.62, 2.1)
	CZP 200	**3.3 (1.99, 5.6)**	-	-
	CZP 400	**3.39 (1.96, 5.94)**	-	-
	ETA 50	**7.12 (3.18, 15.62)**	-	-
	ETA 100	**4.8 (2.08, 10.84)**	-	-
	GOL 50	**3.06 (1.61, 5.61)**	-	-
	GOL 100	1.85 (1, 3.25)	-	-
	INF	1.21 (0.66, 2.12)	-	-
	ABT	**13.97 (7.96, 26.01)**	1.19 (0.54, 2.59)	1.55 (0.79, 3.15)
	APR 30	**10.2 (5.99, 16.9)**	1.78 (0.98, 3.15)	**1.96 (1.2, 3.38)**
	TOF 5	**5.55 (3.48, 9.12)**	1.09 (0.56, 2.14)	1.5 (0.89, 2.6)
	Placebo	**27.49 (10.19, 89.9)**	**1.96 (1.22, 3.14)**	**2.43 (1.65, 3.68)**
IXE Q4W	IXE Q2W	0.85 (0.56, 1.27)	0.73 (0.41, 1.25)	1.16 (0.52, 2.55)
	SEC 150	**1.52 (1.1, 2.1)**	0.7 (0.4, 1.19)	1.65 (0.79, 3.23)
	SEC 300	0.98 (0.72, 1.34)	0.62 (0.37, 1.03)	1.01 (0.48, 2.02)
	GUS Q8W	**0.53 (0.37, 0.76)**	0.56 (0.3, 1.02)	1.29 (0.67, 2.44)
	TIL Q12W	**2.63 (1.35, 5.41)**	-	-
	UST 45	1.41 (0.97, 2.05)	1 (0.49, 1.99)	2.15 (0.98, 4.36)
	UST 90	1.14 (0.8, 1.61)	0.72 (0.36, 1.43)	1.91 (0.87, 3.89)
	ADA	**2.21 (1.7, 2.87)**	0.75 (0.48, 1.17)	1.61 (0.82, 3.19)
	CZP 200	**2.4 (1.48, 4.04)**	-	-
	CZP 400	**2.47 (1.45, 4.29)**	-	-
	ETA 50	**5.21 (2.34, 11.38)**	-	-
	ETA 100	**3.49 (1.54, 7.79)**	-	-
	GOL 50	**2.24 (1.21, 3.95)**	-	-
	GOL 100	1.35 (0.74, 2.32)	-	-
	INF	0.88 (0.49, 1.49)	-	-
	ABT	**10.19 (5.88, 18.58)**	0.96 (0.44, 2.05)	2.24 (0.98, 4.95)
	APR 30	**7.42 (4.49, 11.96)**	1.44 (0.8, 2.51)	**2.84 (1.5, 5.34)**
	TOF 5	**4.05 (2.6, 6.45)**	0.88 (0.47, 1.65)	**2.17 (1.05, 4.3)**
	Placebo	**20.01 (7.55, 65.12)**	1.58 (1, 2.47)	**3.51 (1.9, 6.35)**
IXE Q2W	SEC 150	**1.79 (1.21, 2.69)**	0.96 (0.54, 1.78)	1.43 (0.62, 3.13)
	SEC 300	1.16 (0.79, 1.75)	0.85 (0.48, 1.56)	0.88 (0.38, 1.93)
	GUS Q8W	**0.63 (0.41, 0.98)**	0.76 (0.41, 1.49)	1.12 (0.53, 2.36)
	TIL Q12W	**3.1 (1.55, 6.76)**	-	-
	UST 45	**1.67 (1.08, 2.61)**	1.37 (0.67, 2.93)	1.86 (0.78, 4.19)

Intervention	Comparator	PASI 90	Resolution of enthesitis	Resolution of dactylitis
	UST 90	1.35 (0.88, 2.04)	0.99 (0.48, 2.09)	1.65 (0.69, 3.76)
	ADA	**2.61 (1.76, 3.9)**	1.03 (0.6, 1.85)	1.4 (0.62, 3.16)
	CZP 200	**2.85 (1.64, 5.07)**	-	-
	CZP 400	**2.93 (1.65, 5.36)**	-	-
	ETA 50	**6.18 (2.77, 13.66)**	-	-
	ETA 100	**4.13 (1.79, 9.4)**	-	-
	GOL 50	**2.64 (1.39, 4.9)**	-	-
	GOL 100	1.6 (0.85, 2.84)	-	-
	INF	1.05 (0.56, 1.82)	-	-
	ABT	**12.03 (6.72, 22.77)**	1.32 (0.6, 2.99)	1.94 (0.78, 4.73)
	APR 30	**8.8 (5.11, 14.73)**	**1.97 (1.08, 3.67)**	**2.46 (1.17, 5.18)**
	TOF 5	**4.81 (2.91, 8.15)**	1.21 (0.62, 2.45)	1.88 (0.83, 4.17)
	Placebo	**23.78 (8.73, 78.15)**	**2.17 (1.32, 3.67)**	**3.04 (1.48, 6.22)**
SEC 150	SEC 300	**0.65 (0.5, 0.84)**	0.89 (0.6, 1.31)	**0.61 (0.4, 0.94)**
	GUS Q8W	**0.35 (0.26, 0.47)**	0.8 (0.47, 1.34)	0.78 (0.5, 1.29)
	TIL Q12W	1.72 (0.91, 3.44)	-	-
	UST 45	0.93 (0.65, 1.31)	1.43 (0.74, 2.66)	1.3 (0.79, 2.13)
	UST 90	0.75 (0.54, 1.03)	1.04 (0.54, 1.92)	1.16 (0.7, 1.9)
	ADA	**1.45 (1.1, 1.93)**	1.08 (0.68, 1.73)	0.98 (0.55, 1.8)
	CZP 200	1.58 (1, 2.56)	-	-
	CZP 400	1.63 (0.99, 2.73)	-	-
	ETA 50	3.43 (1.54, 7.5)	-	-
	ETA 100	**2.3 (1, 5.17)**	-	-
	GOL 50	1.47 (0.79, 2.64)	-	-
	GOL 100	0.89 (0.49, 1.54)	-	-
	INF	**0.58 (0.33, 0.98)**	-	-
	ABT	**6.72 (3.99, 11.98)**	1.38 (0.68, 2.77)	1.36 (0.71, 2.7)
	APR 30	**4.89 (2.98, 7.84)**	**2.06 (1.25, 3.31)**	**1.73 (1.09, 2.84)**
	TOF 5	**2.68 (1.77, 4.11)**	1.26 (0.71, 2.25)	1.32 (0.8, 2.21)
	Placebo	**13.18 (5.07, 42.22)**	**2.27 (1.6, 3.17)**	**2.13 (1.5, 3.1)**
SEC 300	GUS Q8W	**0.54 (0.4, 0.73)**	0.9 (0.54, 1.51)	1.28 (0.8, 2.13)
	TIL Q12W	**2.67 (1.42, 5.37)**	-	-
	UST 45	**1.44 (1.02, 2.03)**	1.61 (0.83, 3.05)	**2.12 (1.27, 3.53)**
	UST 90	1.16 (0.82, 1.6)	1.16 (0.61, 2.18)	**1.88 (1.12, 3.16)**
	ADA	**2.25 (1.74, 2.88)**	1.21 (0.81, 1.85)	1.6 (0.89, 2.96)
	CZP 200	**2.45 (1.55, 3.98)**	-	-
	CZP 400	**2.52 (1.53, 4.23)**	-	-
	ETA 50	**5.28 (2.38, 11.54)**	-	-
	ETA 100	**3.55 (1.57, 7.89)**	-	-
	GOL 50	**2.27 (1.22, 4.09)**	-	-
	GOL 100	1.37 (0.75, 2.4)	-	-
	INF	0.9 (0.49, 1.53)	-	-
	ABT	**10.37 (6.16, 18.56)**	1.55 (0.76, 3.14)	**2.22 (1.14, 4.43)**
	APR 30	**7.56 (4.57, 12.12)**	**2.31 (1.41, 3.74)**	**2.82 (1.76, 4.72)**
	TOF 5	**4.12 (2.75, 6.37)**	1.41 (0.81, 2.53)	**2.15 (1.29, 3.66)**
	Placebo	**20.34 (7.85, 65.25)**	**2.54 (1.81, 3.6)**	**3.48 (2.39, 5.17)**

Intervention	Comparator	PASI 90	Resolution of enthesitis	Resolution of dactylitis
GUS Q8W	TIL Q12W	**4.96 (2.63, 9.85)**	-	-
	UST 45	**2.68 (1.79, 3.93)**	1.79 (0.86, 3.66)	1.66 (0.95, 2.74)
	UST 90	**2.15 (1.46, 3.11)**	1.3 (0.62, 2.61)	1.48 (0.84, 2.45)
	ADA	**4.17 (3.04, 5.74)**	1.35 (0.78, 2.35)	1.25 (0.71, 2.19)
	CZP 200	**4.56 (2.86, 7.42)**	-	-
	CZP 400	**4.69 (2.8, 7.91)**	-	-
	ETA 50	**9.87 (4.18, 22.31)**	-	-
	ETA 100	**6.63 (2.77, 15.46)**	-	-
	GOL 50	**4.24 (2.12, 7.99)**	-	-
	GOL 100	**2.57 (1.31, 4.74)**	-	-
	INF	1.67 (0.87, 3.04)	-	-
	ABT	**19.28 (11.06, 35.2)**	1.73 (0.81, 3.65)	1.74 (0.9, 3.3)
	APR 30	**14.11 (8, 24)**	**2.58 (1.49, 4.35)**	**2.2 (1.45, 3.38)**
	TOF 5	**7.7 (4.95, 11.96)**	1.58 (0.84, 2.97)	**1.68 (1, 2.76)**
	Placebo	**37.63 (14.58, 123.93)**	**2.84 (1.88, 4.26)**	**2.72 (1.93, 3.81)**
TIL Q12W	UST 45	0.54 (0.25, 1.06)	-	-
	UST 90	**0.43 (0.2, 0.85)**	-	-
	ADA	0.84 (0.42, 1.6)	-	-
	CZP 200	0.91 (0.42, 1.94)	-	-
	CZP 400	0.94 (0.42, 2.01)	-	-
	ETA 50	1.99 (0.68, 5.21)	-	-
	ETA 100	1.33 (0.45, 3.62)	-	-
	GOL 50	0.85 (0.33, 1.99)	-	-
	GOL 100	0.51 (0.2, 1.18)	-	-
	INF	**0.34 (0.13, 0.76)**	-	-
	ABT	**3.91 (1.65, 8.86)**	-	-
	APR 30	**2.84 (1.22, 6.13)**	-	-
	TOF 5	1.55 (0.72, 3.13)	-	-
	Placebo	**7.57 (2.47, 28)**	-	-
UST 45	UST 90	0.8 (0.57, 1.14)	0.72 (0.43, 1.23)	0.89 (0.55, 1.44)
	ADA	**1.56 (1.11, 2.24)**	0.75 (0.39, 1.48)	0.76 (0.41, 1.44)
	CZP 200	**1.7 (1.03, 2.91)**	-	-
	CZP 400	**1.75 (1, 3.09)**	-	-
	ETA 50	**3.67 (1.68, 7.95)**	-	-
	ETA 100	**2.46 (1.1, 5.54)**	-	-
	GOL 50	1.58 (0.87, 2.79)	-	-
	GOL 100	0.96 (0.53, 1.64)	-	-
	INF	0.62 (0.35, 1.04)	-	-
	ABT	**7.21 (4.15, 13.26)**	0.97 (0.43, 2.16)	1.04 (0.53, 2.15)
	APR 30	**5.26 (3.19, 8.43)**	1.44 (0.75, 2.76)	1.33 (0.81, 2.33)
	TOF 5	**2.87 (1.8, 4.7)**	0.88 (0.43, 1.84)	1.01 (0.59, 1.78)
	Placebo	**14.17 (5.25, 45.61)**	1.59 (0.91, 2.8)	**1.64 (1.09, 2.55)**
UST 90	ADA	**1.94 (1.4, 2.73)**	1.04 (0.55, 2.03)	0.85 (0.46, 1.63)
	CZP 200	**2.12 (1.3, 3.58)**	-	-
	CZP 400	**2.18 (1.28, 3.81)**	-	-
	ETA 50	**4.57 (2.14, 9.7)**	-	-

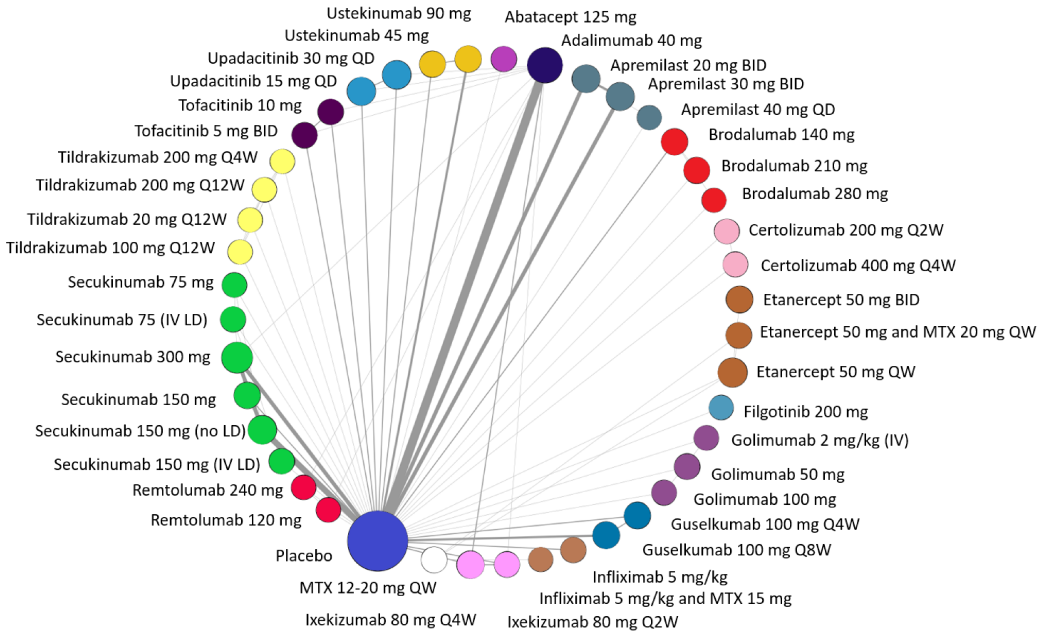
Intervention	Comparator	PASI 90	Resolution of enthesitis	Resolution of dactylitis
	ETA 100	**3.06 (1.39, 6.72)**	-	-
	GOL 50	**1.97 (1.11, 3.37)**	-	-
	GOL 100	1.19 (0.69, 1.97)	-	-
	INF	0.78 (0.45, 1.26)	-	-
	ABT	**8.96 (5.28, 16.28)**	1.34 (0.6, 2.97)	1.18 (0.59, 2.42)
	APR 30	**6.53 (4.1, 10.31)**	**1.99 (1.05, 3.8)**	1.49 (0.9, 2.63)
	TOF 5	**3.57 (2.28, 5.77)**	1.22 (0.6, 2.54)	1.14 (0.66, 2.01)
	Placebo	**17.73 (6.6, 56.27)**	**2.2 (1.28, 3.83)**	**1.84 (1.22, 2.88)**
ADA	CZP 200	1.09 (0.68, 1.77)	-	-
	CZP 400	1.12 (0.67, 1.89)	-	-
	ETA 50	**2.36 (1.07, 5.11)**	-	-
	ETA 100	1.58 (0.68, 3.51)	-	-
	GOL 50	1.02 (0.55, 1.79)	-	-
	GOL 100	0.61 (0.33, 1.05)	-	-
	INF	**0.4 (0.22, 0.67)**	-	-
	ABT	**4.62 (2.71, 8.33)**	1.28 (0.61, 2.62)	1.38 (0.66, 2.9)
	APR 30	**3.38 (2.03, 5.34)**	**1.91 (1.13, 3.15)**	**1.76 (1.01, 3.08)**
	TOF 5	**1.84 (1.21, 2.85)**	1.17 (0.67, 2.05)	1.34 (0.75, 2.38)
	Placebo	**9.06 (3.5, 28.99)**	**2.1 (1.43, 3.06)**	**2.17 (1.33, 3.52)**
CZP 200	CZP 400	1.03 (0.56, 1.89)	-	-
	ETA 50	2.15 (0.9, 5.17)	-	-
	ETA 100	1.45 (0.58, 3.59)	-	-
	GOL 50	0.93 (0.44, 1.85)	-	-
	GOL 100	0.56 (0.27, 1.09)	-	-
	INF	**0.37 (0.18, 0.7)**	-	-
	ABT	**4.25 (2.17, 8.41)**	-	-
	APR 30	**3.08 (1.62, 5.6)**	-	-
	TOF 5	1.69 (0.96, 3)	-	-
	Placebo	**8.31 (2.98, 27.92)**	-	-
CZP 400	ETA 50	2.09 (0.85, 5.13)	-	-
	ETA 100	1.41 (0.56, 3.54)	-	-
	GOL 50	0.9 (0.42, 1.86)	-	-
	GOL 100	0.54 (0.25, 1.1)	-	-
	INF	**0.36 (0.17, 0.71)**	-	-
	ABT	**4.12 (2.1, 8.46)**	-	-
	APR 30	**3.01 (1.53, 5.65)**	-	-
	TOF 5	1.64 (0.91, 2.99)	-	-
	Placebo	**8.09 (2.84, 27.88)**	-	-
ETA 50	ETA 100	**0.67 (0.48, 0.94)**	-	-
	GOL 50	**0.43 (0.19, 0.94)**	-	-
	GOL 100	**0.26 (0.11, 0.56)**	-	-
	INF	**0.17 (0.08, 0.36)**	-	-
	ABT	1.98 (0.82, 4.8)	-	-
	APR 30	1.42 (0.66, 3.07)	-	-
	TOF 5	0.78 (0.34, 1.86)	-	-
	Placebo	**3.92 (1.17, 14.03)**	-	-

Intervention	Comparator	PASI 90	Resolution of enthesitis	Resolution of dactylitis
ETA 100	GOL 50	0.64 (0.27, 1.46)	-	-
	GOL 100	**0.39 (0.17, 0.88)**	-	-
	INF	**0.25 (0.11, 0.56)**	-	-
	ABT	**2.95 (1.18, 7.38)**	-	-
	APR 30	2.13 (0.95, 4.78)	-	-
	TOF 5	1.16 (0.48, 2.82)	-	-
	Placebo	**5.83 (1.68, 21.66)**	-	-
GOL 50	GOL 100	0.6 (0.36, 1.01)	-	-
	INF	**0.39 (0.22, 0.71)**	-	-
	ABT	**4.59 (2.27, 9.78)**	-	-
	APR 30	**3.32 (1.87, 6.05)**	-	-
	TOF 5	1.82 (0.94, 3.71)	-	-
	Placebo	**9.16 (2.95, 30.67)**	-	-
GOL 100	INF	0.65 (0.37, 1.15)	-	-
	ABT	**7.57 (3.83, 16)**	-	-
	APR 30	**5.47 (3.14, 9.83)**	-	-
	TOF 5	**3.02 (1.56, 6.08)**	-	-
	Placebo	**15.06 (4.98, 50.38)**	-	-
INF	ABT	**11.6 (5.9, 24.36)**	-	-
	APR 30	**8.42 (4.88, 14.98)**	-	-
	TOF 5	**4.61 (2.48, 9.19)**	-	-
	Placebo	**23.29 (7.69, 76.41)**	-	-
ABT	APR 30	0.73 (0.37, 1.36)	1.49 (0.73, 3.04)	1.27 (0.67, 2.43)
	TOF 5	**0.4 (0.21, 0.75)**	0.91 (0.42, 2.02)	0.97 (0.48, 1.92)
	Placebo	1.97 (0.68, 6.51)	1.64 (0.87, 3.1)	1.57 (0.87, 2.81)
APR 30	TOF 5	0.55 (0.31, 1)	0.61 (0.34, 1.13)	0.76 (0.45, 1.24)
	Placebo	2.73 (0.96, 8.67)	1.1 (0.77, 1.59)	1.23 (0.88, 1.7)
TOF 5	Placebo	**4.92 (1.8, 16.37)**	**1.8 (1.09, 2.93)**	**1.62 (1.08, 2.44)**

Bolded values indicate a statistically significant difference.

ABT, abatacept; ADA, adalimumab 40 mg Q2W; bDMARD, biologic disease modifying anti-rheumatic drug; BID, twice daily; BRO 210, brodalumab 210 mg; CZP 200, certolizumab pegol 200 mg; CZP 400, certolizumab 400 mg; ETA 50, etanercept 50 mg per week; ETA 100, etanercept 100 mg per week; GOL 50, golimumab 50 mg; GOL 100, golimumab 100 mg; GUS, guselkumab 100 mg; INF, infliximab 5 mg/kg; IXE, ixekizumab 80 mg; PASI, Psoriasis Area and Severity Index; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks; SEC 150, secukinumab 150 mg; SEC 300, secukinumab 300 mg; TIL, tildrakizumab 100 mg; TOF 5, tofacitinib 5 mg BID; UST 45, ustekinumab 45 mg; UST 90, ustekinumab 90 mg

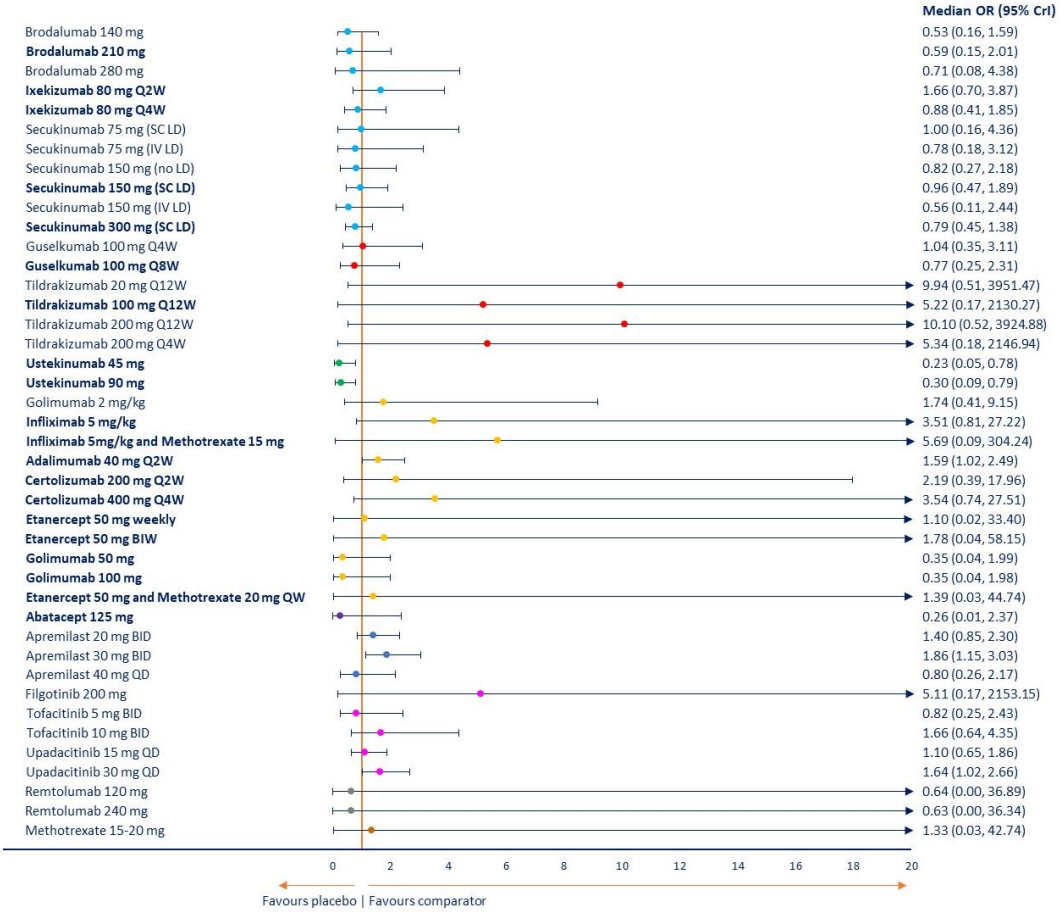
Figure S 14 Network diagram for discontinuation due to AEs



Node size denotes total number of patients on that treatment; line thickness denotes total number of studies informing that comparison.

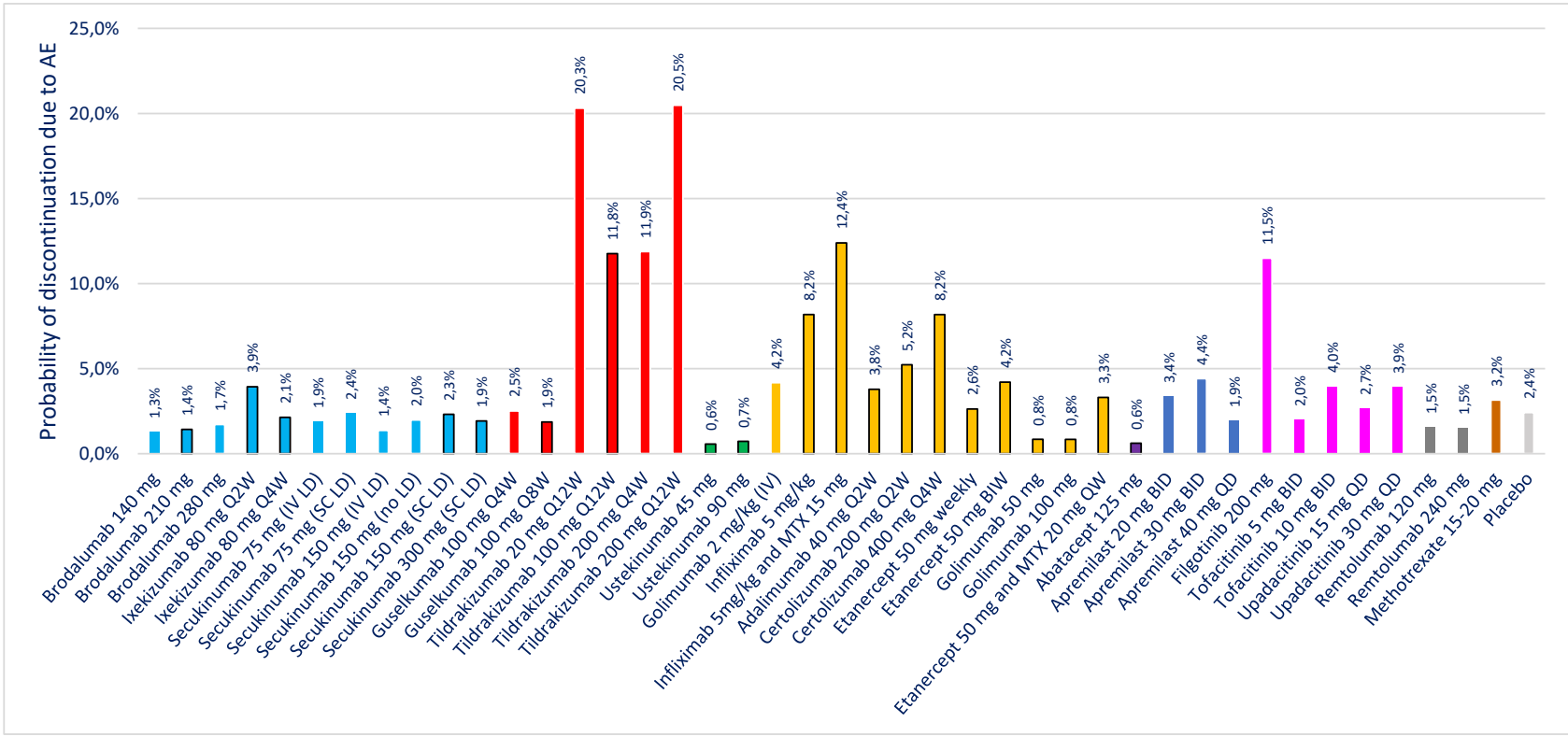
Abbreviations: BID, twice daily; BIW, twice weekly; IV, intravenous; LD, loading dose; MTX, methotrexate; mg, milligram; mg/kg, milligram per kilogram; QD, daily; QW, weekly; Q2W, every 2 weeks; Q4W, every 4 weeks; Q8W, every 8 weeks; Q12W, every 12 weeks.

Figure S 15 Forest plot of treatment effects for all interventions versus placebo on discontinuation due to AEs



DAE treatment effect was based on a fixed-effects model without placebo adjustment. Key comparators are shown in bold.
Abbreviations: BID, twice daily; BIW, twice weekly; CrI, credible interval; DAE, discontinuation due to adverse events; IV, intravenous; kg, kilogram; LD, loading dose; mg, milligram; MTX, methotrexate; QD, daily; QW, weekly; Q2W, every two weeks; Q4W, every four weeks; Q8W, every eight weeks; Q12W, every twelve weeks.

Figure S 16 Probability of discontinuation due to adverse events for all interventions



Probability of discontinuation due to adverse events was based on a fixed-effects model without placebo adjustment. Key comparator bars are outlined in black.
Abbreviations: BID, twice daily; BIW, twice weekly; CrI, credible interval; IV, intravenous; kg, kilogram; LD, loading dose; mg, milligram; MTX, methotrexate; SUCRA, surface under the cumulative ranking curve; QD, daily; QW, weekly; Q2W, every two weeks; Q4W, every four weeks; Q8W, every eight weeks; Q12W, every twelve weeks

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